

**MARINER PARTNERSHIP – VALIDATION AND DEMONSTRATION OF A
RELIABLE, EFFICIENT, SCALABLE, AND LOW-COST PEM FUEL CELL SYSTEM**

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1. Excellence #@REL-EVA-RE@#

1.1 Objectives and ambition #@PRJ-OBJ-PO@#

1.1.1 Overall Objective

Global shipping represents around 90% of global trade¹ and with more than 99% using fossil fuels² constituting approximately 3% of total global emissions. If left unchecked, maritime CO₂ alone could grow 50–250 % by 2050³. To change course, the EU has placed the sector firmly under the **Fit-for-55** package:

- **EU-ETS maritime** came into force on 1 Jan 2024 and will move to full cost coverage by 2026; it already prices 50 % of extra-EU and 100 % of intra-EU voyages.
- The **FuelEU-Maritime Regulation** sets tightening GHG-intensity limits that reach –80 % in 2050, effectively banning fossil-only propulsion⁴.

At the same time, the international **Mission Innovation “Clean Hydrogen Mission”** and its European counterpart, the **Clean Hydrogen Joint Undertaking (CH JU)**, have adopted converging targets: make clean H₂ available at ≤ **2 USD/kg by 2030** and deploy **100 Hydrogen Valleys** worldwide. Within that framework, the CH JU SRIA defines explicit **Key Performance Indicators (KPIs) for maritime PEM fuel-cell (FC) systems by 2030**:

KPI	Target 2030	Strategic Bottleneck	MARINER Outcome
Rated power ≥ 10 MW		Multi-stack scaling & modular BoP	Scalable 1 MW module validated; electrical & fluid interfaces sized for 10 MW replication.
Lifetime 80 000 h		Diagnostics, prognostics, health-aware control	SL-AST × 40 acceleration + digital twin extrapolation to 60 000 h demo, 80 000 h design.
Cost (€/kW) ≤ 1000		Standardisation, volume manufacturing	1 MW prototype at ≤ 1 200 €/kW (2028) → learning-curve model to 1 000 €/kW (2030).

Our project is conceived as the **missing link between high-TRL 2–3 MW demonstrators (e.g. HyShip, HyEkoTank) and a commercially viable product platform** that ship designers can specify as off-the-shelf by the 2027–2030 delivery window. MARINER will:

- **Engineer and validate a 1 MW PEMFC system as a modular “building block”**—fully aligned with the *StasHH* module standard—ready to be paralleled to ≥ 10 MW.
- **Demonstrate ≥ 1 000 h in moist-saline, variable-temperature conditions** that reflect engine-room and on-deck placements, and **accelerate a 200 kW slice for 40 000 h equivalent** via a newly harmonised System-Level Accelerated Stress Test (SL-AST).
- **Cut CAPEX toward 1 200 €/kW in 2028 and on trajectory to 1 000 €/kW by 2030** through standardised power-conversion “bricks”, replicated BoP skids and digital-twin-driven over-design reduction.
- **Embed health-conscious control and fault-tolerant EMS** so vessels can maintain power even with partial stack or DC/DC failures—crucial for IMO safety-return-to-port rules.
- **Quantify total-cost-of-ownership (TCO) and life-cycle footprint** for two end-user routes (tanker & high-speed ferry) and benchmark against LNG, batteries and ammonia ICE, proving compliance with the EU *Do-No-Significant-Harm* principle.

By aligning with these KPIs, MARINER de-risks the first wave of ETS-exposed retrofits and establishes a European technology and supply-chain leadership position in the global race to decarbonize deep-sea shipping.

Strategic relevance:

- **Climate impact** Replacing a 10 MW diesel genset (typical Handymax tanker hotel load) with green-H₂ PEMFC at 55 % LHV avoids ~35 000t CO₂ over 20 years⁵. Scaling to 5 000 vessels (5 % of world fleet) yields ~175 Mt CO₂ cumulated savings → 3 % of the EU’s 2050 net-zero gap.
- **Industrial sovereignty** The project consolidates EU supply chains for stacks, power electronics and control software, mitigating over-reliance on Asian imports.
- **Standardization & regulation** Close cooperation with Lloyd’s Register and the StaSHH and ZEWT partnerships ensures that outputs feed directly into IMO MSC and CEN/CENELEC rule-making, clearing the path to serial class approval.

In short, MARINER turns the Clean Hydrogen Mission’s aspirational cost and performance goals into a concrete, qualification-ready product platform. By building directly on recent state of the art PEMFC wins generated by Clean

¹ International Chamber of Shipping, *Shipping and World Trade – Facts & Figures*

² IMO 4-th GHG Study 2020, Module 1

³ European Parliamentary Research Service, *Reducing shipping’s climate impact* (2020)

⁴ Regulation (EU) 2023/1805 (FuelEU Maritime), Art 4

⁵ Calculated by MARINER consortium from EEDI reference curves & 630 g CO₂/kWh MGO baseline

Hydrogen JU projects, focussing the technical objectives to the maritime KPIs, and further, by embedding open-source models, harmonised JRC test protocols, and a TRL-9 road-map into the workplan, MARINER becomes a direct execution vehicle for the Clean Hydrogen Mission: **it converts stack-level breakthroughs into a deployable, certifiable, economically-viable power-train family, accelerating the point where zero-emission shipping is the rational business choice, not an R&D pilot.** That is the strategic canvas against which the specific, measurable objectives that follow (Section 1.1.2) have been defined.

1.1.2 Specific, Measurable Objectives

SO1. Low-Cost, Modular Multi-Stack PEMFC “Building-Block” for Maritime Use	
- In WP3 GENEVOS will collaborate with CEA, VARD, MINES, and LR EMEA to develop a general-purpose multi-stack PEMFC system for maritime applications capable of achieving MW-scale power output, with a roadmap towards type-approval by a classification society. - This system will serve as a flexible building block for larger systems up to 10 MW, ensuring optimal cost, efficiency, durability, and reliability across various vessel types and operational profiles. This includes defining the optimal stack arrangement and power rating of each module, as well as the system architecture including air, hydrogen, cooling, and power management subsystems, to achieve scalability, modularity and cost reductions. - Increase system efficiency and lifetime through innovative design that pioneers new heat recovery techniques, maximises fuel conversion, and provides variable control of stack temperature, pressure, and humidity. - Explore power architectures (DC/DC) with a high level of modularity in electronics, mechanics, and control that is optimized for the PEMFC modular system. This will be developed with a view to facilitating scalability to meet the power levels required for maritime applications. - Implement advanced control strategies to enhance system performance and ensure continuous operation even if individual stacks fail, thereby increasing overall reliability and reducing the need for costly, vessel-specific designs. - Demonstrate the durability of the multi-stack system in relevant environmental conditions and use cases.	
Specific Objectives	Objectives KPIs
Develop a fuel cell module with a continuous power rating in the range 100-300 kW	KPI 1.1 - Compliant with SatSHH standards KPI 1.2 - CAPEX < 1000 EUR/kW
Develop a 1 MW multi-stack system based on StaSHH standards.	KPI 1.3 - Marine compliance and roadmap towards type-approval KPI 1.4 - 1 MW continuous operation at 55 % LHV system efficiency KPI 1.5 - > 60000 h of system lifetime KPI 1.6 - CAPEX < 1200 EUR/kW
SO2. Advanced Simulation, Monitoring & Control Tools for Multi-MW PEMFC Systems	
- Develop and integrate advanced PEMFC models (digital twins) that consider degradation aspects at both the cell and system levels. These models will help in optimising system design, improve efficiency, and extend the durability of fuel cell systems. - Utilize model-based diagnostics to inform health-conscious control strategies, covering both lower level BoP control such as water management and, and Energy Management System optimization of load-sharing between fuel cell modules and battery. These will include both physics-based and data-driven (AI) methods, such as anomaly detection with auto-encoders. - Integrate the developed models and methodologies into open-source tools like VirtualFCS to facilitate their adoption in other sectors, such as heavy-duty road transportation and rail, thereby amplifying the impact beyond maritime applications.	
Specific Objectives	Objectives KPIs
Develop and validate advanced models enabling fast simulation of PEMFC behavior, including degradation aspects	KPI2.1 - RMSE < 10 mV for beginning-of-life polarization curve and load-cycle simulation KPI2.2 - RMSE < 20 mV for EOL polarization curve and load-cycle simulation KPI2.3 - Load-cycle simulation > 100x faster than real-time
Implement developed PEMFC models in an open-source system-level simulation tool	KPI2.4 - Implementation of two developed models in VirtualFCS

Development of advanced control algorithms for multi-stack fuel cell systems	KPI2.5 - Lifetime at system level: 80.000 hours
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SO3. Durability Validation in Real Maritime Conditions via System-Level AST

- Develop system-level Accelerated Stress Testing (AST) protocols that accurately represent maritime operating conditions and dynamic load requests. These protocols will be used to evaluate the reliability and durability of high-power fuel cell modules over 40,000 hours.
- Calibrate fuel cell degradation models using AST data to improve protocol definition and extend assessment of system performances, lifetime and scalability over a wide range of use cases.

Specific Objectives	Objectives KPIs
Define a fuel cell system level AST protocol for maritime applications	KPI3.1 - Validation with at least 1000h in relevant environmental conditions (moistly, salty, different air inlet temperature) KPI3.2 - > 20x acceleration of fuel cell system degradation
Development of data driven aging models fed with AST results	KPI3.3 - Extrapolation of degradation for more than 40000 hours KPI3.4 - Fuel cell system lifetime estimation for different use cases.

SO4. Provide Performance, Safety & Sustainability Benchmarks Across Vessel Segments

- Conduct Total Cost of Ownership (TCO) and Life Cycle Analysis (LCA) for two use cases chosen with end users. Compare the multi-stack PEMFC solution with alternative powertrain options such as batteries, biofuels, and Solid Oxide Fuel Cells (SOFC).
- Evaluate the system safety and reliability and produce procedures, recommendations and best practices for end users on multi-MW FCS.
- Integrate TCO and LCA models with fuel cell system simulations to predict key performance indicators early in the design phase, enabling joint optimization of technical, economic, and environmental factors. This approach will significantly reduce development costs by avoiding the need for bespoke systems for each vessel type.

Specific Objectives	Objectives KPIs
Develop a tool to calculate TCO and perform LCA for a given system design and load profile	KPI4.1 - Publicly available tool is released and feedback received
Establish a comparison between PEMFC and alternative powertrain solutions	KPI4.2 - Compare TCO and GHG emissions for at least two segments (offshore and cruise) and two alternatives (biofuels, batteries, ammonia ICE or SOFC).
Identify regulatory compliance pathway and assess risks of modular arrangement of fuel cell power system	KPI4.3 - Develop safety procedures, recommendations and best practices for end-users

SO5. Accelerate Industry Uptake Through Publications, Best-Practice Guides and Targeted Training of End-Users

- Widely disseminate the project's findings, get feedback and promote the adoption of the developed multi-stack PEMFC system across the maritime industry and beyond.
- Engage with industry stakeholders, publish peer-reviewed articles, and present at major international conferences to showcase the system's benefits and encourage its integration into maritime operations.

Specific Objectives	Objectives KPIs
Publish project results in academic journals and conferences	KPI5.1 - Publish at least 6 papers in peer-reviewed journals KPI5.2 - Participate in > 10 international conferences
Provide training support and build capacity of end-users on FC degradation modelling and control	KPI5.3 - 1 session organised KPI5.4 - > 10 participants trained
Engage with Industry stakeholders and European Standardisation organisations to support future market uptake	KPI5.6- Presentation of project results in at least 5 European working groups (incl. standardisation technical committees)

SO6. Strengthen EU Supply-Chain and Cross-Project Synergies via Multi-Stakeholder Seminars and Shared Datasets and Protocols

- Coordinate with other related projects, such as H2MARINE, StaSHH, HyShip, FLAGSHIPS, MARANDA, and ShipFC, to leverage synergies, avoid duplication of efforts, and ensure complementarity.
- Establish a collaboration framework with these projects to share knowledge, best practices, and resources, fostering a unified approach to advancing fuel cell technology in the maritime sector.

Specific Objectives	Objectives KPIs
Establish annual seminars with partners from related projects EU projects around the development of FC for maritime applications	KPI6.1 - Organize at least two seminars with > 30 people KPI6.2 - Cross-sector partnerships established (e.g., with other EU-funded projects, industry consortia, or clusters): ≥ 5 partnerships

1.1.3 Ambition and Advance Beyond State-of-the-Art

To achieve these ambitious goals, the MARINER partnership proposes a holistic approach that is organized in two interconnected Technological Pillars (TP), aimed at enabling affordable and reliable PEMFCs: **TP1 - Modular, scalable and affordable FCS architecture**; **TP2 - Simulation, monitoring and control for extreme durability**; composed of 6 Key technology Areas (KTAs; see Section 1.1.3), which in turn give rise to **13 Specific Project Results (R)**, all embedded into a work plan (see section 3.1).

TP1 – MODULAR, SCALABLE AND AFFORDABLE FCS ARCHITECTURE

KTA 1. Standardized fuel cell module configuration for scalability and cost reduction

SoA: Current maritime PEMFC systems often comprise a single large stack or non-standard multi-stack setups. This can increase complexity in manifolding and balancing, leading to elevated costs and reliability concerns⁶. Existing projects (e.g., HyShip) show promise but lack standardized modular solutions that facilitate straightforward scaling to multi-megawatt levels. Recently, the European consortium StaSHH⁷ proposed an open standard for fuel cell stack module size, physical interfaces and communication in heavy-duty applications. Further, sea air pollutants like NaCl, SO₂, and NO_x can degrade PEMFC performance and components⁸. Effective air filtering is crucial, but designing and maintaining these systems in marine environments is challenging. Research suggests adapting operations or cleaning procedures⁹ to mitigate effects. A robust testing protocol is needed to characterize filter performance and aging, guiding predictive maintenance and optimization.

Progress Beyond:

1. **Modular, standardized design** supporting up to 10 MW by validated multi-stack+BOP duplication, minimizing redesign efforts.
2. **Redundancy and fault-tolerant operation** for improved uptime in maritime applications.
3. **Reduced CAPEX through common stack frames**, shared BoP elements, and usage of widely available components.
4. **Class validation and FAT at module level** reducing certification and test costs associated with fuel cell power installation

Positioning (TRL 4 → TRL 6): We build on lab-scale stack modules (TRL 4) and demonstrate a 1 MW multi-stack architecture in relevant conditions (up to TRL 6). We rely on a patented design allowing for a gas-tight separation of electrical components and the fuel cell system, avoiding ATEX components and thus lowering costs (pending patents US2024204229A1, JP2024514629 (A), CN117355968A, EP4327385A1 by GENEVOs).

KTA 2. Highly-efficient, scalable, multi-stack system for multi-MW maritime applications

SoA: A challenge for low temperature PEM fuels is significant energy losses through heat generation and dissipation, as well as high energy consumption on balance of plant, and suboptimal conditions for humidity, which is difficult to control through the use of passive membrane humidifiers. BoP are often specific to the overall systems to date which require bespoke designs and components, increasing cost, maintenance access and preventing easy power upgrades. Existing solutions also typically do not address advanced cooling, oxygen supply, and water management under highly variable load profiles¹⁰. Further, optimization of system design is often based on efficiency, rather than durability, or else focus on cell-level durability improvements¹¹.

⁶ Gadducci *et al.* BoP incidence on a 240 kW PEMFC system in a ship-like environment, employing a dedicated fuel cell stack model. *Int. J. Hydrogen Energy* **46**, 24305–24317 (2021).

⁷ <https://stashh.eu/>

⁸ Broer *et al.* Polymer electrolyte membrane fuel cell degradation in ships — Review of degradation mechanisms and research gaps. *J. Power Sources* **640**, 236678 (2025).

⁹ Briand, A. *et al.* Effects of sodium chloride on PEMFC durability at a concentration close to that of a marine environment. *J. Electrochem. Soc.* **171**, 104508 (2024); Lamard *et al.* Impact of NaCl spray on the durability of PEMFC single cells and stacks in marine environment. *J. Electrochem. Soc.* **170**, 024504 (2023).

¹⁰ Gadducci *et al.* BoP incidence on a 240 kW PEMFC system in a ship-like environment, employing a dedicated fuel cell stack model. *Int. J. Hydrogen Energy* **46**, 24305–24317 (2021).

¹¹ Cullen, D. A. *et al.* New roads and challenges for fuel cells in heavy-duty transportation. *Nature Energy* **6**, 462–474 (2021).

Progress Beyond:

1. **Highly efficient, scalable BoP architecture, optimized for extended durability**, applicable both at module, multi-stack (e.g. compressors, humidification) and multi-module levels (e.g. air and exhaust treatment).
2. **Increased heat recovery** to improve net efficiency, through variable stack operating temperature and new methods of precisely controlling humidification at different power levels to improve performance and lifetime
3. **Full hydrogen fuel recovery** to increase efficiency and prevent purged, unused fuel often associated with injectors and recirculation. Use of new methods and development of hardware to convert 100% of hydrogen fuel, ensuring that no hydrogen is purged which results in energy loss and associated secondary greenhouse gas emissions.

Positioning: (TRL 4 → TRL 6): A pilot-scale BoP (TRL 4) will be adapted and tested to maritime specifications, culminating in a fully integrated 1 MW system (TRL 6).

KTA 3. Optimized modular DC/DC power-conversion architecture

SoA: Traditional architectures such as series, parallel, cascade, or series-parallel connections are well-established, they each come with limitations¹². The parallel architecture, for instance, offers the highest level of modularity, enabling individual control of each fuel cell stack. However, it imposes significant constraints on DC/DC converter components, potentially reducing reliability, and fails to achieve the efficiency levels of the series architecture. The optimal architecture depends on the application, and an appropriate selection can lead to significant gains in terms of efficiency and lifetime.¹³ Further, in today's MW-scale marine PEMFC pilots the DC/DC stage is almost always a one-off, bespoke converter sized for the whole stack string (series architecture). This brings (i) high CAPEX (> 150 €/kW for power electronics), (ii) limited redundancy and (iii) prolonged type-approval because the hardware is non-standard. Commercial, marine-qualified "brick" converters (250–400 kW, 600–1500 VDC) do exist, but there is no validated design guide on how to assemble them into a fault-tolerant, multi-MW architecture for maritime applications.

Progress Beyond:

1. Identification of **state-of-the-art marine-grade off-the-shelf DC/DC modules**, to lower development costs and ensure scalability.
2. **Optimization methodology for designing a power-conversion architecture** based on advanced fuel cell and degradation models.
3. Validation of the proposed architectures with a **hardware-in-the-loop (HiL) platform allowing easy configuration and prototyping of the different power conversion stages**, the different controllers (FCMS, PMS, EMS) and communications.

Positioning (TRL 4 → TRL 6): The proposed HiL platform and the demonstration of a 1 MW system will allow the test of electrical architectures and their components in a relevant environment for maritime applications (TRL6).

TP2 – SIMULATION, CONTROL AND MONITORING FOR EXTREME DURABILITY

KTA 4. Fuel Cell Data-Driven and Physics-Based Models for Durability Analysis

SoA: Current modeling approaches for PEMFCs in maritime contexts are limited in their ability to accurately capture degradation phenomena under more severe environmental constraints. Most models used in system design or control are either too simplified (neglecting aging) and/or computationally heavy, limiting their real-time application. Recent works presented control-oriented physics-based degradation models¹⁴ with a 0D approach allowing for fast simulations. However, they lack the extensive validation and maturity required for a reliable application in the system design process and control applications. Moreover, PEM fuel cell models are rarely available as open source, which prevents reuse or extension of PEM fuel cell models, model comparison and benchmarking¹⁵.

Progress Beyond: A first contribution of this project is to further validate state-of-the-art physics-based degradation models for the particular conditions of maritime applications. This project also introduces hybrid modeling frameworks that combine physics-based degradation mechanisms and performance models with data-driven insights derived from extensive operational datasets and Accelerated Stress Tests (AST). These models will be used to create high-fidelity digital twins enabling:

1. **Real-time degradation simulation** to support design, validation and predictive diagnostics;

¹² Zhou, S. *et al.* A review on proton exchange membrane multi-stack fuel cell systems: architecture, performance, and power management. *Appl. Energy* **310**, 118555 (2022).

¹³ Igourzal *et al.* Improving the operation of multi-stack PEM fuel cell systems to increase efficiency, durability and lifespan while reducing ageing. *Energy Convers. Manag.* **314**, 118630 (2024).

¹⁴ Touil *et al.* Control oriented modeling of integrated catalyst and membrane degradation in PEM fuel cells. in *2024 IEEE Transportation Electrification Conference and Expo (ITEC)* 1–7 (IEEE, 2024).

¹⁵ Vetter & Schumacher. Free open reference implementation of a two-phase PEM fuel cell model. *Comput. Phys. Commun.* **234**, 223–234 (2019).

2. **Accurate long-term performance assessments** across various maritime use-cases; and
 3. **Open-source integration**, encouraging easy and broad adoption among different maritime end-users.
- Model validation will leverage experimental data from consortium partners (MINES, CEA, SEC). The developed models will be made available in an existing open-source simulation platform.

Positioning (TRL 4 → TRL 6):

This task will start from validated lab-scale models of single cells or fuel stacks (TRL 4), and extend them to full-system, real-time capable tools, validated against AST for maritime applications and available in an open-source simulation platform (TRL 6).

KTA 5. Fuel Cell Stack SoH Monitoring and Control

SoA: SoH estimation in current maritime PEMFC systems is limited to basic voltage monitoring or conservative rule-based diagnostics¹⁶. Similarly, control strategies in existing systems are typically pre-defined, with key parameters such as recirculation rates and purge events often based on fixed maps or worst-case assumptions¹⁷. Without precise SoH tracking and corresponding control laws, current systems risk suboptimal operation, premature maintenance, or unexpected failures.

Progress Beyond: MARINER will implement a **multi-level SoH monitoring framework** enabling:

1. **Early detection of aging or performance degradation** at the module and stack level, helped by Machine Learning methods; and,
2. **SoH-aware control strategies**, including adaptive cooling and humidity level to maximize lifetime and maintain safe operation.

In addition to SoH estimation, this KTA introduces advanced control strategies specifically tailored to maritime PEMFC systems, including: (i) dynamic pressure control of air and hydrogen and integrated control of auxiliary systems (e.g., humidifiers, pumps), to ensure optimal and smooth operation for each stack (ii) recirculation control to manage hydrogen utilization and improve efficiency, while mitigating water accumulation and membrane drying ; and (iii) purge strategy optimization, to maximize autonomy.

Positioning (TRL 4 → TRL 6): Building on lab-scale prognostics and control algorithms (TRL 4), this KTA will implement validated SoH monitoring and advanced control strategies on the integrated 1 MW system under real conditions (TRL 6), demonstrating operational benefit capabilities.

KTA 6. Multi Scale EMS for Independent Module/Stack Control

SoA: Energy management systems (EMS) in current maritime applications typically control PEMFC as a monolithic block, even in the case of a multi-unit integration, with limited flexibility to adapt power flow at the level of individual modules or stacks^{18,19}. The power demand is generally distributed proportionally between each module and this heuristic strategy consistently lead to sub-optimal efficiency and elevated aging²⁰, particularly when one module underperforms²¹.

Progress Beyond: MARINER will introduce a multi-scale Fuel-Cell Management System, capable of:

1. **Independently controlling each module's power output**, thermal management, and safety status;
2. **Managing failure cases by dynamically reallocating power**, ensuring partial operation even during faults; and
3. **Optimizing load sharing** based on real-time SoH data and predicted degradation.

This enables flexibility and system modularity and availability, especially for long-distance missions.

Positioning (TRL 4 → TRL 6): This task will start with modular EMS control strategies developed in lab environments (TRL 4), and demonstrate a robust, fault-tolerant multi-MW EMS framework deployed and tested at system level in a relevant maritime scenario (TRL 6).

¹⁶ Li *et al.* Health state monitoring and predicting of proton exchange membrane fuel cells: A review, in *Journal of Power Sources*, Vol. 612, August 2024; Vanem *et al.* Data-Driven Approaches to Diagnostics and State of Health Monitoring of Maritime Battery Systems, in *Annual Conference of the PHM Society*, October 2023

¹⁷ Steinberger *et al.* Anode recirculation and purge strategies for PEM fuel cell operation with diluted hydrogen feed gas, in *Applied Energy*, Vol. 232, December 2018

¹⁸ Shi *et al.* Multiple Time Scale Energy Management for a Fuel Cell Ship Propulsion System, in *Proc of the 2024 European Control Conference*, 2024

¹⁹ Shi, J., Aarsnes, U. J. F., Tao, S., Wang, R., Nærheim, D., & Moura, S. (2023). Health-Aware Energy Management for Multiple Stack Hydrogen Fuel Cell and Battery Hybrid Systems. *Applied Energy*, to appear.

²⁰ Igourzal *et al.* Improving the operation of multi-stack PEM fuel cell systems to increase efficiency, durability and lifespan while reducing ageing. *Energy Convers. Manag.* **314**, 118630 (2024).

²¹ Marx *et al.* A review of multi-stack and modular fuel cell systems: Interests, application areas and on-going research activities, in *International Journal of Hydrogen Energy*, 2014

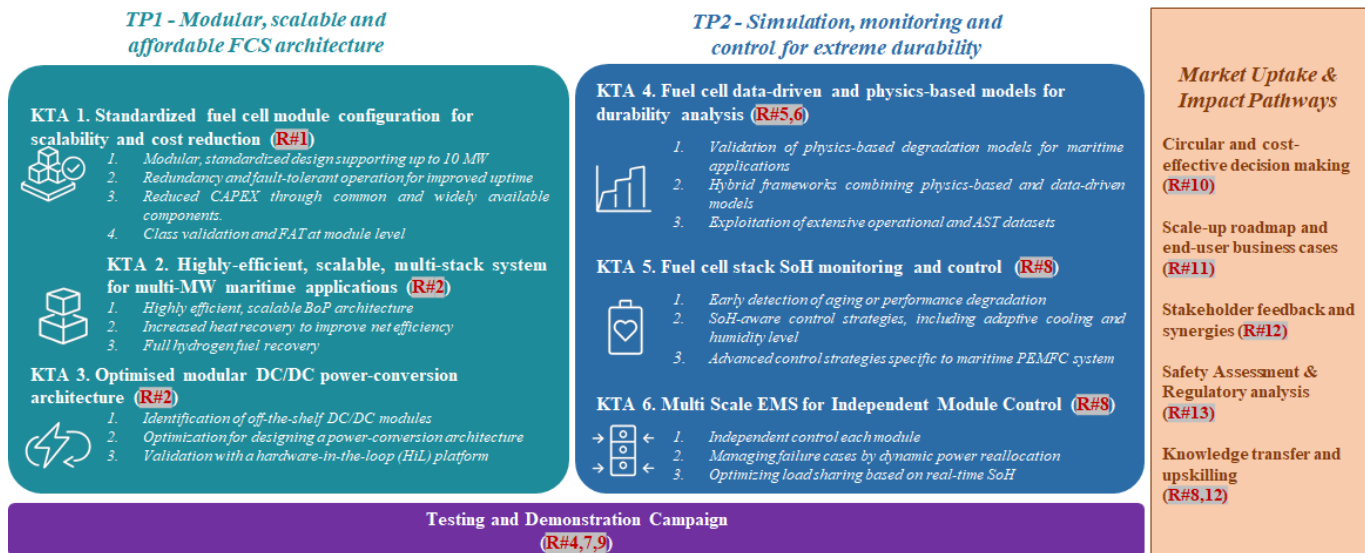


Figure 1: Overall concept of the MARINER project.

#§PRJ-OBJ-PO§#

1.2 Methodology #@CON-MET-CM@# #@COM-PL-CP@#

1.2.1 Overall Methodological Approach (~ 5 page)

The MARINER project will be structured around the development of the 1 MW multi-stack fuel cell system, which corresponds to its first technological pillar. As such, the first task will be to identify target vessel types and load cycles, collect operating data and requirements from end-users in order to define system specifications (T3.1 - Design requirements definition). This phase will see a tight cooperation between representative end-users (KIARA, SCORPIO, MAERSK TANKERS) the system integrator (VARD) and FCS OEM (GENEVOS), with support from a classification society (LR EMEA). We will then move to the definition of the stacks, balance of plant and the overall system architecture, which will be based on unity fuel cell modules (T3.3 - System architecture & Conceptual Design). This task will be led by GENEVOS but will be supported by the early characterization of performance and durability of short-stacks and of the air filtering system (T3.2 - Subsystem pretesting of concepts). The electrical architecture will be the object of particular attention, with the optimization of power electronics based on advanced fuel cell models and the validation under relevant conditions in a HiL platform (T3.4 - Optimized electrical architecture design for the modular system). Once a first design is defined, a ~ 200 kW fuel cell system module will be built (T4.1 - Component Procurement & Assembly 200 kW FCS). This first FCS at a smaller scale will allow us to evaluate the concept, perform initial tests and start accelerated stress tests. It will allow us also to implement design-based safety checks and best practices for maritime operation. Feedback and lessons learned from this first system will allow us to fine tune the design of the 200 kW module (T4.2 - 200 kW FCS Validation testing & development) and complete the detailed design of the 1 MW system (T3.5 - Detailed FCS design), while incorporating maritime reliability constraints, single/multiple failure modes & safety measures into the design (T3.6 - Reliability (Design Phase)). System modules will be then validated prior to full system integration (T4.4 - Modular FCS Validation / FAT) and the 1 MW FCS will be built (T4.3 - Assembly 1 MW FCS).

Two core tests will allow us to demonstrate the performance and durability of our 1+ MW FCS: accelerated stress tests at the module scale and real conditions tests at the full 1 MW scale. At the unit module scale (~200 kW), we will perform an accelerated stress test in which we expect to demonstrate the durability of the FCS module under relevant conditions. The first challenge will be to define representative AST protocols for maritime applications. We will rely on a literature review, physics-based fuel cell degradation models and short-stack tests performed in WP3 to define and validate an appropriate AST protocol (T5.1 - Extended-AST protocol definition). We will then run the system-level AST with the selected protocol and extrapolate results with the aim of data-driven aging models fed with AST results (T5.2 - Module level AST implementation and analysis) to assess FCS durability for different maritime applications considering system upscale (T5.3). At the 1 MW scale, we will setup (T6.1 - Setup, installation and decommissioning) and run 1 MW FCS endurance tests in a testing platform from SEC aiming to reproduce relevant environmental and operating conditions (T6.2 - 1 MW FCS endurance test ($\geq 1,000$ h)). These endurance tests will be based on a simulated EMS which aims to provide the fuel cell with the load demands corresponding to different use profiles. Following both tests we will liaise with JRC to contribute to the definition of appropriate test protocols and benchmarks for maritime applications (T6.3 - Harmonized testing protocols for

performance and durability assessment).

For achieving the 80000 h FCS lifetime objective, the MARINER project will rely on its second technological pillar: Simulation, control and monitoring for extreme durability. The FCS design will be supported by fuel cell system simulation capabilities, in order to ensure an optimal compromise between cost, efficiency, durability and reliability. Therefore, in parallel, we will work on the integration of advanced fuel cell models (including durability aspects) into simulation platforms used by GENEVOS for fuel cell system design. Fuel cell models, both physics-based and data-driven, have been previously developed by MINES, CEA, USTUTT, but also by GENEVOS. In this project, we intend to bring those models to the next level by sharing the expertise acquired by those partners and to confront our models to further validation, with a focus on operating conditions and materials relevant for maritime applications (T7.1 - Multi-MW Simulation Software). We will also leverage on the durability data produced by the AST tests at both short-stack and FCS module scales. That will develop data-driven and/or hybrid degradation models in order to extrapolate the results for 1 MW and durability for different use cases (T5.3 - System level AST results extrapolation and durability assessment). We also aim to develop tools to estimate the TCO (T8.1 - TCO Assessment for 1 MW & 10 MW) and perform an LCA analysis (T8.2 Circularity & LCA). The goal is to have key economical and environmental indicators as early as possible in the design phase, so that they can be taken into account in critical choices. The second technological pillar also involves the development of monitoring algorithms able to evaluate the state-of-health of the system (T7.2 Monitoring Algorithms) and of control algorithms which will ensure a health-conscious operation of the multi-stack FCS (T7.3 Control Algorithms). The different fuel cell models, monitoring and control algorithms and TCO/LCA tools will be published as open-source code in order to benefit from feedback from the community and increase the impact of the project (T7.4 Open Source Publishing).

Reliability and safety are key factors for marine fuel cell systems. In the MARINER project, they will be ensured by the long standing experience of Lloyd's Register (LR EMEA), combined with the technical expertise of a system integrator (VARD) and of a FCS OEM (GENEVOS). During the design phase, we will be particularly attentive to the reliability and operational safety, identifying single/multiple failure modes and incorporating safety measures into design (T3.6 - Reliability (Design Phase)). That will be supported by a review of regulatory aspects and gaps aiming to identify compliance pathways (T10.1 - Review of Regulatory aspects and Gap analysis) and a risk assessment including different incident/accident and reliability scenarios at both module (T10.2 - Risk assessment of fuel cell power system development) and full system scales (T10.3 - Risk assessment of modular arrangement of fuel cell power system). We will perform a safety review and develop procedures, recommendations, and best practices for end users (T10.4 - Safety analysis and Recommendations).

All the efforts of the MARINER project will be driven towards the scalability, technological maturity and adoption of the developed fuel cell system. The results of the TCO and LCA analyses will be translated into cost/improvement guidelines (T8.3 - Benchmark & Techno-Economic Recommendations). Along with lessons learned throughout the project activities, they will also serve as input to a scale-up feasibility study towards the 10 MW target (T9.1 - Scale-Up Feasibility (1 MW → 10 MW)). A strategy to increase TRL from 6 to 9 will be defined, with at least one specific end-user scenario in collaboration with a ship operator (T9.2 - Roadmap to TRL9 & Use Case), while identifying potential synergies and eventual modifications in architecture to address sectors other than the maritime, such as road or stationary applications (T9.3 - Investigate Spillover). We will be particularly attentive to gather user input and feedback at regulatory and non-technical level, with two yearly meetings with the Stakeholder advisory board and the organization of ad hoc workshops (T9.4 - Advisory Board & User Engagement: Feedback and Best Practices). Ensuring knowledge exchange and maintaining links with other relevant projects (H2MARINE, ZEWT, StaSHH, HyShip, etc.) will also be important for us, and will be the subject of a specific task (T9.5 - External Partnerships & Synergies) led by CERTH, who also participates in the H2MARINE.

TP1 – MODULAR, SCALABLE AND AFFORDABLE FCS ARCHITECTURE

KTA 1. Standardized fuel cell module configuration for scalability and cost reduction

We will focus on developing a standardized, modular fuel cell system architecture that supports scalability up to 10 MW, addressing the unique challenges of maritime environments. This approach begins with a comprehensive requirements and concept study, involving detailed analysis of end-user needs, technology review, and alignment with marine safety regulations, and research on existing fuel cell system designs. The goal is to define a modular architecture that minimizes complexity and cost, adhering to the StaSHH open standard for stack module size, physical interfaces, and communication protocols. Subsystem pre-testing is conducted to characterize critical components like the fuel cell stack and air filtering system, which will drive the multi-stack and system sizing for determining the final module size and BOP arrangement that allows for the highest cost reduction at scale, whilst also ensuring redundancy.

KTA 2. Highly efficient, scalable, multi-stack system for multi-MW maritime applications

High net efficiencies and increased durability and stack lifetime are dependent on significant development and innovation of the fuel cell system balance of plant, which comprises the air, hydrogen, cooling and control loops, together with the multi-stack arrangement that it manages at module level. Firstly, we will perform a literature review, combined with consortium wide brainstorming and workshops, together with gathering key initial supplier data, to allow the development and definition of several concepts of system architecture. For these we will derive simplified models to provide a high-level evaluation and comparison of net system efficiency, lifetime and cost. We will then analyze leading system concepts and expand the detail of modeling using a physics-based numerical model of the FCS. Short-stack performance and degradation characterisation testing as well as key supplier info will feed into this to calibrate the model and determine the target operating conditions of the fuel cell stack. Next, we include degradation processes in the model to allow us to consider the impact of operating conditions on efficiency and durability early in the design phase. This enables us to drive the design and development of the BOP system architecture, through an optimisation process that will seek to maximize efficiency and lifetime, with respect to the performance KPI's, whilst meeting the system cost and scale up criteria.

A core part of the design innovation will focus on maximising heat recovery and full conversion of H₂ to increase electrical efficiency and increasing the level of variable control of the stack operating parameters (humidity, temperature, pressure, startup sequence) in order that optimal conditions can be met for the key system operating points. It is important to note that efficiency and degradation test data from the 200kW prototype will be leveraged to develop and confirm the detailed design of the 1 MW system, ensuring scalability and integration of improvements. Component and subassembly validation testing are conducted to de-risk full-scale testing and reduce debugging time, leading to the full assembly of the 200 kW FCS prototype. Validation testing of the 200 kW FCS includes functional, performance, and safety-related assessments, with iterative software and hardware upgrades to optimize system performance. Finally, running a system-level accelerated stress tests on the 200 kW FCS prototype will allow us to demonstrate its durability performance and provide feedback to refine the design of the 1 MW system.

KTa 3. Optimised modular DC/DC power-conversion architecture

Our first step involves conducting a thorough market analysis and technical evaluation to identify state-of-the-art marine-grade off-the-shelf DC/DC modules. This process will select components that meet the required specifications for maritime applications, ensuring they are robust, reliable, and cost-effective. Following this, an optimization methodology will be developed for designing a power-conversion architecture based on the physics-based fuel cell performance and degradation model previously mentioned, calibrated from short-stack characterisation data. This methodology will leverage previous work by MINES and CEA and allow us to consider various operational scenarios, load profiles, and degradation factors specific to maritime environments, aiming to balance efficiency, durability, reliability, and cost-effectiveness.

Next, a hardware-in-the-loop (HiL) platform will be used for easy configuration and prototyping of different power conversion stages, controllers (FCMS, PMS, EMS), and communication protocols. This platform will simulate real-world conditions and enable the testing of various electrical architectures in a relevant environment (humidity and salty conditions), ensuring that the proposed designs meet performance and reliability requirements. The HiL platform will be used to validate the proposed electrical architectures through extensive testing, assessing their performance under various load conditions, and evaluating their fault-tolerance and redundancy capabilities. Finally, the validated electrical architecture will be demonstrated in a 1 MW system, showcasing its performance and reliability in a relevant maritime environment.

TP2 – SIMULATION, CONTROL AND MONITORING FOR EXTREME DURABILITY

KTa 4. Fuel cell data-driven and physics-based models for durability analysis

Using fuel cell models is key to extrapolate AST results to other scales and use cases. Physics-based models have the advantage of offering insights into the physical phenomena occurring in the fuel cells, as well as predictive capabilities for design optimization and diagnostics. However, they can be computationally intensive and complex to develop and parametrize. Data-driven models provide improved accuracy, rapid development, and real-time applicability, but they rely heavily on the quality and quantity of available data and may struggle with extrapolation

beyond training conditions. In this project we will work with physics-based models and a hybrid approach allowing us to make the most of both physics-based and data-driven models.

First, we will adopt an advanced 0D physics-based model with very low computational requirements (~ 2 ms to simulate a 10-point polarization curve), which is being developed by MINES. Such a very fast model allows for the simulation of long operation times (> 1000 h), the integration in more complex system-level models, and for extensive parameter estimation and validation. Short-stack tests will be used to calibrate and validate this model so that it is available very early to support system design and the definition of relevant AST protocols. The model will also be validated against data from the 200kW module AST and the 1 MW system endurance tests.

By using data-driven surrogate models, combining physics-based models and machine learning algorithms fed with data coming from AST in a hybrid approach, we will develop an FCS digital twin. The digital twin should simulate with the greatest precision its behavior subjected to extreme conditions and help to extrapolate the results of accelerated stress testing at system scale in terms of durability. The digital twin will also enable the analysis of degradation mechanisms, the identification of critical parameters influencing system lifespan, and the prediction of long-term performance beyond the limits of physical testing. Extrapolating experimental data, the digital twin will be used to adjust testing protocols and provide outcomes to optimize wear scenarios, reducing the need for prolonged physical trials, and to enhance the reliability of durability forecast.

KT A 5. Fuel cell stack SoH monitoring and control

MARINER will implement a multi-level SoH monitoring framework using both physical measurements (voltage, pressures, temperature, flows) and data-driven analytics. To anticipate potential degradation while keeping improving performance, a particular attention will be given to the monitoring of the FC membranes humidity level, which are not directly measurable and should be recovered by alternative means.

These strategies will enable:

- **early detection of aging or performance degradation** at the module and stack level, helped by Machine Learning methods.
- **SoH-aware control strategies**, including adaptive cooling and humidity level to maximize lifetime and maintain safe operation.

In addition to SoH estimation, this KTA introduces advanced control strategies specifically tailored to maritime PEMFC systems, and depending on SoH estimation. This includes dynamic pressure control of air and hydrogen and integrated control of auxiliary systems (e.g., humidifiers, pumps) based on both system state and predicted degradation trends, to ensure optimal and smooth operation for each stack. A second point of study will deal with recirculation control to manage hydrogen utilization and improve efficiency, while mitigating water accumulation and membrane drying. A particular attention will be given on purge strategy optimization, particularly for nitrogen and water management, using real-time sensor feedback and SoH estimates to trigger purging only when necessary—thus minimizing hydrogen loss and improving autonomy.

These advanced control algorithms will be validated under real and accelerated conditions, enabling lifetime extension and durability of the FC systems. Besides, test campaigns and AST results will feed into previously designed simulation softwares to validate SoH estimation techniques for various degradation modes.

KT A 6. Multi Scale EMS for Independent Module/Stack Control

Energy management systems (EMS) in current maritime PEMFC architectures typically control the system as a monolithic block, with limited flexibility to adapt power flow at the level of individual stacks. This static management strategy leads to energy losses, particularly when one module underperforms due to a degraded SoH, for instance.

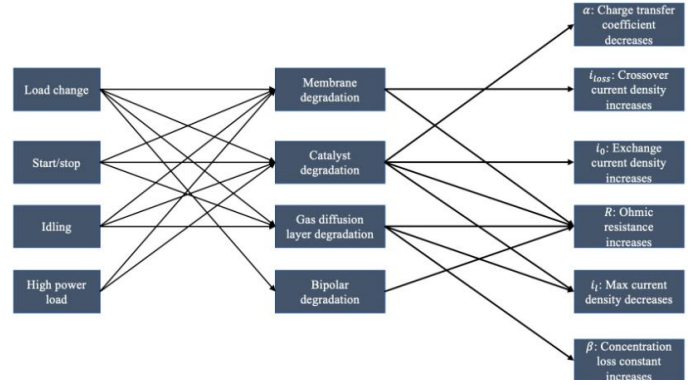


Figure 2: Fuel cell degradation caused by four operational conditions.

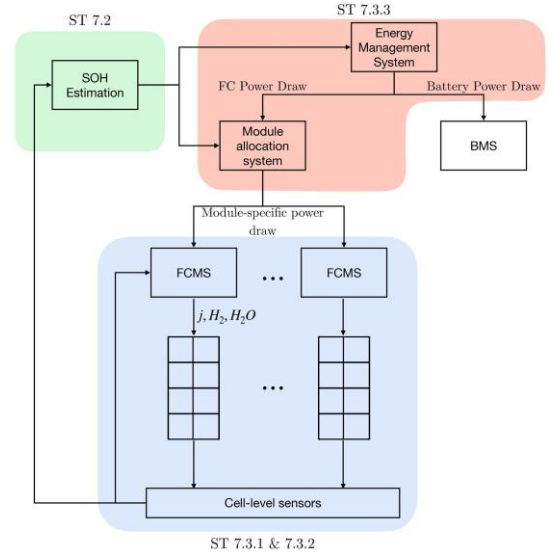


Figure 3: Schematic of the proposed multi-scale control architecture.

MARINER will introduce a multi-scale Fuel-Cell Management System, designed to adapt dynamically to the Modules/Stacks monitoring. At the top level, this Management System will distribute the power demand between the battery management system and the FC. At the FC level, this FCMS will independently control each module's power output, thermal management, and safety status and optimize load sharing based on real-time SoH data and predicted degradation. In addition, it will manage failure cases by dynamically reallocating power, ensuring partial operation even during faults.

This approach will thus enable unmatched flexibility, modularity, and system availability, especially for long-distance or mission-critical vessels. It will be tested on a Hardware-in-the-Loop (HiL) testing platform at CEA, specifically designed for prototyping modular fuel cell architectures.

TESTING AND DEMONSTRATION CAMPAIGN FOR COMPONENT AND SYSTEM VALIDATION, AND GAUGING DEGRADATION RATES UNDER REALISTIC MARITIME CONDITIONS

To achieve Technology Readiness Levels (TRL) 6 and ensure progression towards TRL 7 and beyond for the innovative solutions developed within MARINER, three laboratory facilities have been selected for pretesting, testing, validating, and demonstrating the PEMFC system performance under realistic maritime conditions. These labs are used in conjunction with architecture design and FC development and follow a step-by-step procedure, from short-stack pre-tests, to submodule AST and finally the test of 1 MW full FC system in realistic conditions. This facilitates systematic assessment of system reliability, efficiency, durability, and degradation rates. The three testing environments, managed by experienced consortium partners, offer complementary capabilities and ensure the testing demands are met at all stages of the project.

TESTLAB #1: Paris (France), MINES

Location: Sophia-Antipolis, France, Lab facility: <10 kW fuel cell test bench, Testing Type: short-stack, sub module pretest, Testing length: 880 hours @ 2kW.

DESCRIPTION OF LAB FACILITY AND TESTING PLAN

Lab overview: The short-stack test bench of Centre PERSEE/Mines Paris is able to test short-stacks and systems up to 10 kW with controlled operating conditions (pressure, temperature, relative humidity). The test bench was designed and built internally, which provides us a high level of flexibility to test different fuel cell system configurations. With four high-humidity relative humidity probes and a condenser for water collection, accurate water balance measurements are possible, providing additional insight into water management issues.

Challenge: Two challenges will be addressed by short-stack tests. The first is to provide the FCS OEM with stack performance and durability data as early as possible in the design process. The second is to define and validate an AST protocol before applying it to the 200 kW FCS module.

Objectives: Characterize the performance of the chosen stack technology, with polarization curves and step response curves for different operating conditions, in order to determine the point of maximal system efficiency. Quantify degradation rates for different operating conditions and for start/stop cycles.

Implementation: Definition of an experimental plan in concertation with other partners (GENEVOS, CEA, etc.). Prepare the test bench and install the stack upon reception. Characterize short-stack beginning-of-life performance (~10 days - 80 h of test, including preliminary tests and debugging). Characterize short-stack load cycle degradation (~ 50 days - 400 h of test). Characterize short-stack AST degradation (~ 50 days - 400 h of test). Experimental data treatment and reporting.

Expected results: Set of polarization and step response curves, characterizing both steady-state and transient performance of the chosen stack technology.

Partners involved: MINES, GENEVOS, CEA



TESTLAB #2: SEANERGY, CEA

Location: Bouguenais, France, **Lab facility:** Marine high-power energy systems test facilities (fuel cells and batteries), **Testing Type:** AST, Simulated environment, **Testing length:** ≥ 1000 hours @ 200 kW.

DESCRIPTION OF LAB FACILITY AND TESTING PLAN

Lab overview: The SEANERGY platform provides a unique infrastructure, dedicated to conception and qualification of new energy systems and materials for maritime applications. It addresses more specifically : 1) durability and reliability in marine environment 2) multi-MW upscaling for maritime sector. Among others, SEANERGY is equipped with a fuel cell test facility in a 55m³ climatic chamber allowing to test FC systems up to 500 kW in specific humidity/temperature conditions as well as with air conditioning on the inlet (e.g. salt, pollutants). It is also equipped with an Hardware-in-the-Loop (HiL) equipment dedicated to electrical architecture prototyping and control validation.

Challenge: Methodology for testing and assessing durability of high-power FCS.

Traditional Accelerated Stress Tests (AST) primarily focus on material-level degradation, employing stress methods and characterization techniques that are not feasible at the system scale due to their intrusive nature. Moreover, the solicitations used in standard AST are not representative of the system use because all the BoP components have an impact on stack operating conditions. This creates a gap between laboratory-based durability assessments and real-world system performance.

Scaling a FC system for maritime applications is almost done by paralleling integration of stand-alone modules, around 100-200 kW, that have undergone specific design, validation and certification. Power-intensive maritime applications encourage more innovative architecture design, enabling modular integration for upscaling while optimizing the upstream (air and H₂ supply) and downstream (electrical integration) architecture. Full-scale physical testing and assessment of such new architecture concepts become then a major challenge for fuel cell systems at MW scale

Objectives: To bridge this gap, our innovative approach introduces system-level accelerated stress testing (SL-AST), designed to replicate operational stresses harsher than real-world and significantly reducing testing time, and modular electrical architecture prototyping to evaluate system control and upscale feasibility.

Implementation: This approach will include e.g. dynamic load variations, hot and cold start-stop cycles, environmental conditions (moistly, salty, different air inlet temperature), multi parameter environmental stress combination to capture interdependent failure modes.

Expected results: Assessing reliability and durability of high-power FCS dedicated to maritime applications, develop and set-up data-driven surrogate models of fuel cell aging using the test results.

Partners involved: CEA, Genevos, Mines, LR EMEA.



TESTLAB #3: Energy House, SEC

Location: Stord, Norway; **Lab facility:** Full scale “ship on land” marine testing facility. **Testing Type:** Full system in simulated environment; **Testing length:** 1.000 hours @ 1000 kW.

DESCRIPTION OF LAB FACILITY AND TESTING PLAN

Lab overview: The Energy House at Sustainable Energy Catapult Centre offers a comprehensive full-scale testing environment tailored for hydrogen-based energy systems and other zero-emission technologies. The lab is equipped with:

- 1 MW green hydrogen production from electrolysis, based on local renewable energy
- Fuel cell testing setups for multi-hundred kW systems
- Battery testing infrastructure and storage capabilities



for both hydrogen and ammonia

- Safety systems and instrumentation for continuous monitoring. This enables system-level validation of components and integrated solutions under real-world operational conditions.

Challenge: Testing of full-scale energy systems such as fuel cell and hybrid systems often lacks realistic and safe environments that combine multiple variables – load cycles, environmental fluctuations, and interaction between subsystems. Laboratory conditions often fail to reflect the complex interplay of components in operational settings, especially when integrating storage, fuel supply and power conversion.

Objectives: To validate the reliability, safety, and performance of a 1 MW fuel cell system under simulated real-world conditions, including:

- Dynamic load profiles resembling maritime/off-grid operations
- Interaction with hydrogen production and storage systems
- Operational testing under Norwegian climatic conditions
- System response during normal operation, failure modes, and grid interactions

Implementation: The test plan includes continuous 1,000-hour operation using green hydrogen, dynamic load cycling, and intermittent startup/shutdown procedures. Data acquisition will capture key parameters including thermal behavior, energy efficiency, emissions, and fault responses. Testing will include black start capabilities, integration with battery systems, and simulation of maritime power demands. The hydrogen price at the test facility is expected to decrease from approximately 150 NOK/kg in 2025 to 105 NOK/kg by 2028, aligning with the planned test period. This projection is based on key factors such as increased production volumes leading to improved economies of scale, access to low-cost and stable energy from local hydropower resources, and continuous optimisation of operations and maintenance practices. Additionally, ongoing technological advancements in electrolysis efficiency and system integration are anticipated to further reduce production and operational costs, supporting a competitive hydrogen price by 2028.

Expected results: Validation of system stability and performance under varying load and environmental conditions, Identification of system bottlenecks and optimization opportunities, Safety and control system performance verification, and Basis for certification and pre-commercial deployment

Partners involved: SEC, GENEVOS, VARD.



PEMFC MODELLING, MONITORING, CONTROL AND OPEN-SOURCE SOFTWARE DEVELOPMENT

An important goal of MARINER is to ensure that developed fuel cell models, monitoring and control algorithms are available in open-source modelling platforms allowing for system-level studies of PEMFC-based systems. To build up on a previous project and an actively developed and maintained tool, we will focus on integrating our developments into the VirtualFCS²² Modelica library.

IMPROVEMENT OF VIRTUALFCS FUEL CELL STACK MODEL

The current model used by VirtualFCS to represent a fuel cell stack is a physics-based 0D model which uses an overly simplified approach where parameters such as the cell exchange current density, crossover current or the ohmic resistance are constant. This simple model cannot capture the response of the fuel cell under different operating conditions (temperature, relative humidity, pressure...) which are known to significantly impact those

²² <https://www.sintef.no/projectweb/virtual-fcs/>

parameters²³. Further, this model does not account for the effect of operating conditions on performance degradation. Those simplifications limit the applicability of VirtualFCS fuel cell stack model in the design process. We intend to add the possibility of using improved fuel cell models in VirtualFCS, pushing forward its simulation capabilities and reliability.

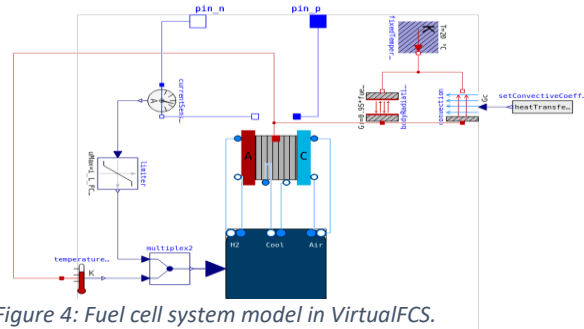


Figure 4: Fuel cell system model in VirtualFCS.

IN DEPTH USE CASE EVALUATION

While the laboratory testing campaign (WP5, WP6) ensures robust technical validation, MARINER's ultimate goal is to prepare the developed multi-MW PEM fuel cell system for real-world maritime deployment and laying the groundwork for subsequent TRL 8 demonstrations. Two complementary use cases, provided by **SCORPIO** and **KIARA**, have been selected to reflect distinct operational profiles, vessel types, and commercial requirements. These use cases will ensure that end-user engagement, regulatory considerations (WP10), and scale-up strategies are informed by real challenges and operational data. The results from these use cases will guide TCO (WP8) and LCA evaluations, enabling a thorough assessment of the system's economic viability and environmental impact in genuine maritime contexts. By validating performance, safety, and practicality in two distinct operational scenarios, MARINER demonstrates its readiness for wider market adoption and provides a strong foundation for multi-MW expansions, aligning with the project's overarching scope and expected outcomes.

USE CASE #1: STI Comandante, SCORPIO

Location: Tallinn-Rotterdam, **Ship Type:** Handymax tanker, load range 400 - 2200 kW, **Operational Profile:** Continuous/Long-Duration.

SHORT DESCRIPTION OF THE USE CASE

Challenge: Scorpio Group operates a diverse fleet with continuous, long-duration voyages in large shipping segments, such as tankers and bulk carriers. Ensuring operational reliability and meeting stringent emissions regulations is critical, especially under stable yet high load demands.

Objectives: [1] Evaluate an integration and degradation with the MARINER 1 MW PEM fuel cell system under prolonged maritime duty cycles. [2] Validate compatibility with existing shipboard power distribution and safety standards.

Expected Results: 1) Quantified economic and environmental impact, informing the roadmap to TRL 9 and multi-MW expansions. 2) Evaluate the business case and enhance stakeholder confidence in hydrogen-based solutions for mainstream commercial shipping.

KPIs: KPI 1.3 - Marine compliance and roadmap towards type-approval; KPI3.4 - Fuel cell system lifetime estimation for different use cases.



USE CASE #2: HSC Super Runner Jet, KIARA

Location: Medium / short range, frequent routes in the Mediterranean (Rafina–Mykonos–Rafina route, a typical Aegean island crossing profile with moderate distances and port infrastructure). **Ship Type:** A high-speed mono-hull ferry (service speeds of 35 knots and a top speed of 40 knots). Overall length of 100.3 meters, powered by four Ruston V20RK 270 engines and KaMeWa 112F11 waterjets, delivering a combined output of 27,500 kW. Capacity for 710 passengers, 120 cars, and 4 buses. **Operational Profile:** Short range ferry, frequent stops.

SHORT DESCRIPTION OF THE USE CASE

²³ Affonso Nóbrega, P. H. A review of physics-based low-temperature proton-exchange membrane fuel cell models for system-level water and thermal management studies. *J. Power Sources* 558, 232585 (2023).

Challenge: KIARA operates high-speed passenger ferries with short, frequent routes, rapid acceleration, and frequent port stops. Achieving reliable transient load response and optimizing efficiency under fluctuating demand is a key challenge. Evaluating safety considerations for passenger ferries, and ensuring the social license to operate will be key.

Objectives: [1] Evaluate the MARINER fuel cell system's dynamic performance under high-load variability. [2] Assess resilience to harsh maritime conditions (salt spray, humidity) and frequent starts/stops. [3] Integrate user feedback (T9.2, T9.4) to refine operational guidelines and regulatory compliance.

Expected results: 1) Evaluation of reliability of operation for high-speed ferries, demonstrating immediate load-following and robust safety measures. 2) Roadmap for integrating fuel cells in short-route passenger shipping. 3) Informed approach for maritime regulations and scaling (WP10), underpinning MARINER's readiness for broader market adoption.

KPIs: KPI 1.3 - Marine compliance and roadmap towards type-approval; KPI3.4 - Fuel cell system lifetime estimation for different use cases.



1.2.2 National or international R&D activities related to the project

The innovations of the MARINER project are built upon past and ongoing national and international R&D initiatives. The consortium partners have extensive collaborative experience from relevant EU- and national-funded projects in the field of hydrogen fuel cell technologies and alternative energy storage solutions for maritime applications. MARINER's consortium will use knowledge and lessons learned from these projects and ensure close coordination with ongoing initiatives by organizing targeted synergy and knowledge-exchange activities (see T9.5). This approach will ensure complementarity, leverage synergies and avoid duplication of efforts. The consortium's positioning in existing projects helps increase the MARINER project impact, facilitate the sharing of best practices, and accelerate innovation through effective alignment with complementary research and industry stakeholders.

H2MARINE | HEU (2024-2027)

Contact with the relevant CETH coordination team

H2MARINE focuses on large-scale maritime stack development. MARINER is the logical continuation: system-level demonstration based on those stacks. It lays the groundwork for marine stack hardware but does not address full system integration. **Planned Actions:** Joint workshop in Year 1 (Task 9.5) with the respective CETH team; Share advisory board members and test plans; Align system design and control interface assumptions with H2MARINE outputs; Include results in simulation and LCA comparisons (WP7/8).

ShipFC | HEU (2020-2025)

NORCE (Former CLARA Venture Labs), SEC

Demonstrates a 2 MW ammonia-fuelled solid oxide fuel cell system on an offshore support vessel. The project provides detailed test data and operational insights from a high-power maritime FC integration. **Planned Actions:** Direct lessons learned integration into testing methodology (T6.1–6.3); Use of SEC's Energy House builds on ShipFC legacy; Use ShipFC data to define use cases and scaling strategy (T9.1, T9.2).

SEABATT | HEU (2021-2025)

VARD, CEA

The overall objective was to develop a full-electric maritime hybrid concept based on combining modular high-energy batteries and high-power batteries. **Planned Actions:** Coordinate on modular integration and hybridization strategies (T3.4, WP7).

V-ACCESS | HEU (2022-2025)

VARD

This project will assess ESS alternatives to batteries, SMES and supercapacitors, and their realistic energy and power density for marine environment to match vessel's operative condition; assess the additional lifetime of batteries when combined to these new ESS; and assess if the different hazards and fire risks allow reduced spaces between individual modules. **Planned Actions:** Share insights on lifetime extension and ESS architecture design for hybrid systems (WP3/7).

H2SEAS | HEU (2024-2026)

GENEVOS

This EU Horizon project proposes a novel, fully-integrated hydrogen-electric small scale fishing vessel to accelerate a sustainable and accessible transition to clean and efficient power for small-scale fishing fleets. The project will implement the design, construction, and operational demonstration of a full hydrogen-electric fishing vessel, with a 80 kW LT PEMFC system, over a duration of 2yrs providing test data to support validation of the fuel cell system

and its durability in the harsh marine environment. **Planned actions:** Leverage long-duration real-sea validation results for input to MARINER's durability in design and control focus (WP5/6).

CLEAN H2 SHIPPING | HEU (2025-2028)

CEA

The ambitious CleanH2shipping project unites ports, research organisations, technology providers and a ship operator, with the aim of developing and implementing a viable hydrogen ecosystem and showcasing the potential of hydrogen fuel-cell-powered inland and short sea shipping. **Planned Action:** Ensure knowledge sharing on safety standards, regulatory protocols, and test data (WP6, WP9, WP10).

StasHH | HEU (2021-2025)

CEA

The EU-funded StasHH project gathers a consortium of 11 fuel cell module suppliers, 9 original equipment manufacturers and 5 research institutes to define, develop and test the first European standard for fuel-cell modules for heavy-duty applications. StasHH will standardise physical dimensions, flow and digital interfaces, and test protocols and safety requirements of fuel cell modules that can be stacked and integrated in heavy-duty applications such as forklifts, buses, trucks, trains, ships, and construction equipment. **Planned Actions:** WP3 will explicitly use StaSHH-compatible module architecture (T3.1–T3.3); Participate in StasHH dissemination through Task 12.1; Design for compliance with StaSHH electrical, cooling, and communication interfaces; Test cross-compatibility of MARINER's FCS modules with other StaSHH-compatible hardware

NEMOSHIP | HEU (2023-2026)

CEA

NEMOSHIP will develop a modular and standardised battery energy storage solution (BESS) that is able to exploit heterogeneous storage units and a cloud-based digital platform to enable data-driven, optimized, and safe exploitation. The project will demonstrate the maturity of these innovations at Technology Readiness Level (TRL) 7 for hybrid ships; the project will also investigate the adaptability towards fully electric ships via semi-virtual demonstration of two additional fully electric vessels employed in ferrying and short-sea shipping. These use cases will focus on the improvement of mid-term modular DC electrical architecture enabling the use of large BESS in combination with other zero-emission power sources like fuel cells. **Planned Actions:** Coordinate on modular architecture and hybrid management (T3.4, T7.3); Joint discussion of modular architectures and electrical safety with LR (WP10).

HyEff | National Project (2022-2025)

NORCE (Coordinator)

Energy efficient operation of hydrogen powered vessels (HyEff) aims to improve energy efficiency and durability of PEM fuel cells and battery hybrid vessels by developing multi-stack, health conscious Energy Management Systems. **Planned Action:** Build on The Health Conscious energy management strategies of HyEff in WP7.

HyValue | National Project (2022-2030)

NORCE (Coordinator)

HyValue is an eight-year Norwegian research center led by NORCE that focuses on hydrogen-based energy carriers with the aim of achieving zero emissions. The research covers hydrogen transport and storage, maritime safety, risk assessment, socio-economic studies, business models, regulatory framework and societal acceptance. **Planned action:** HyValue's work in hydrogen production pathways, and cross-cutting socio-technical analyses will inform the end user use cases developed in MARINER.

OCEANCHARGER | National Project (2022-2025)

VARD, NORCE, SEC

Project's main goal is to develop and demonstrate an offshore charging solution for ships, as well as to describe the scaling up and commercialization of solutions for emission-free vessel operations in offshore wind farms to help accelerate the green transition in the maritime sector. The project includes a full scale test of the solution on a vessel including connection at sea and integration into the power management system. Solution fits for both AC and DC switchboards. **Planned action:** Share system integration insights for multi-MW electrical architectures (T3.4).

DuraSys-PAC | National Project (2023-2026)

MINES, CEA

The French ANR DuraSys-PAC project aims to improve the durability of PEM fuel cells for heavy-duty mobility. It proposes to work on two main axes: (i) the development of a modeling platform to estimate PEMFC lifetime, and (ii) the development of strategies to manage and reduce PEMFC degradation. In that context, we seek to simulate the operation (and degradation) of fuel cells over >10000 h to understand the impact of control strategies on degradation, as the fuel cell state-of-health evolves. **Planned Actions:** Import degradation modeling framework and simulation experience (T5.1–5.3); Use DuraSys-PAC methodologies to validate MARINER AST protocol and hybrid models (T7.1).

HYSySPEM | National Project (2022-2027)

CEA

The French ANR HYSySPEM project aims at improving hybrid PEM fuel cell systems for high-power applications (including ships) with a systemic approach. The optimization of electrical and fluidic architectures in a modular multi-stack approach is carried out at several scales (FCS and hybrid system) and on key components (compressor, power converter at voltage levels from 800V to 1500V). Improvements in local control and power management at

hybridization level are coupled with a fault-tolerant control approach for power converters and hybridization to ensure a very high level of availability. **Planned Action:** Exchange on fluidic/electrical architectures and fault-tolerant control (WP3, WP7, WP10).

1.2.3 Interdisciplinary Aspects and Integration

MARINER is deliberately built around the value-chain needed to cover every competence to take multi-MW maritime PEM-fuel-cell system from concept to type-approval and market uptake. The project relies on a multi- and inter-disciplinary consortium with proven expertise that covers all key required disciplines, from fundamental research and experiments on Fuel Cell degradation pathways (CEA, MINES) to practical expertise in Fuel Cell design and manufacture (GENEVOS). VARD has an impressive track record in integrating multi-MW power generators and batteries with the electrical systems of maritime vessels, required to work under the harshest marine conditions. For demonstration we combine two leading testlabs specialized in large scale maritime FCS tests (CEA, SEC). All development is furthermore overseen and supported by LR EMEA - a leading provider of classification and compliance services to the marine and offshore industries - to ensure readiness for a marine environment, stringent safety and reliability requirements, and a clear roadmap to type-approval. On the software side, the project will leverage the scientific excellence of MINES, with a track record of pushing the cutting edge of modelling fuel cells, and both the theory and application of control and monitoring algorithms. This expertise is combined with USTUTT who has decades of experience building and implementing critical software for the powerhouse that is the German automotive industry. As such, this project provides a unique opportunity to bring the peak of European engineering excellence from the automotive industry into productionalizing software for maritime FCS, and delivering the results as open source. NORCE (including former CLARA VENTURE LABS) is ideally situated with an extensive experience and network within european maritime industry, and with a proven technical expertise in fuel cell system development, to work with end users towards a realistic and actionable roadmap for TRL 9 and a market ready system. This is strategically combined with CERTH and KIARA who brings with them what we consider to be **Europe's best use case for a large deployment of hydrogen powered marine vessels:** fast ferries between the greek islands (20+ vessels at ~10-20 MW, with fixed routes, and lengths of 1-5 hours). This is a strategically important use case for Greece, as it is central for the nations tourist industry and faces a particular pressure to decarbonise. This is also emphasized through the involvement of a representative from the **Greek Ministry of Maritime Affairs and Insular Policy** in the AB. To ensure the developed system is also applicable beyond ferry use cases, and serves the wider maritime market, the consortium also includes SCORPIO and the consultancy NJORD, specializing in retrofitting existing vessels with emerging green energy solutions.

Stakeholder engagement

The project methodology relies on the exchange of knowledge and good practices of many different stakeholders/target groups (**see impact section**): OEM, integrators, shipbuilders and ship operators to confirm the needs and as future direct users and beneficiaries of the MARINER solutions; policy makers, classification organizations and standardisation committees to ensure a safety and standard-compliant design and de-risk scalability and future deployments; RDI projects/initiatives and academic/research institutions to build on the most recent research works; as well as any others stakeholders interested in sharing experience and learning from one another.

For each of the activity where stakeholder engagement will be needed during the project, the following methodology will be applied:

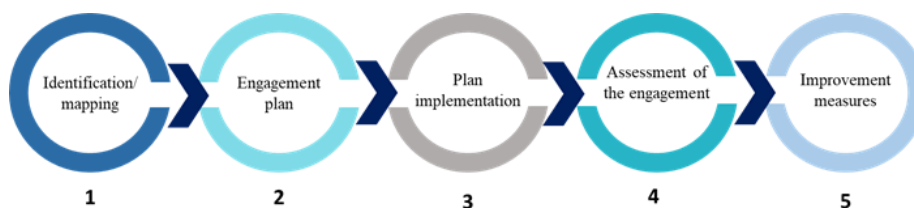


Figure 5: Stakeholder engagement methodology.

This methodology will be refined at the start of the project, with a developed engagement plan for each type of stakeholder detailed in the “Communication, Dissemination and Exploitation Plan” in M6. The engagement plan will define, for each stakeholder group, a potential selection process. Furthermore, **a specific task and budget** has been reserved in WP9 to manage the AB, to gather users feedback through workshops and consultations, to facilitate the communication and keep the members engaged.

At the proposal stage, the stakeholder identification has already started, with the main stakeholders identified being the following ones. **Most of the stakeholders identified so far are part of the ecosystem of MARINER partners.**

Among the potential AB members, **Ficantieri group, Trasmapi and the Greek Ministry of Maritime Affairs and Insular Policy** have already confirmed their participation. They are categorised according to their role in the project.

1.2.4 Gender Dimension in R&I Content

According to the International Maritime Organization (IMO), women make up only about 2% of the global seafaring workforce, and the numbers are similarly low in shipbuilding, and other technical and operational roles.

MARINER will ensure the participation of women across stakeholder engagement, training and dissemination activities. This includes outreach to women's maritime or hydrogen associations (e.g. WITSA International; Women in Green Hydrogen), inclusive representation in panels and engagement with female students and professionals. The project also acknowledges that sex and gender considerations can influence technology usability, safety, and stakeholder engagement. While the fuel cell system itself is gender-neutral, attention will be paid to ergonomic design, maintenance accessibility, and safety procedures to accommodate diverse physical characteristics and operational needs.

1.2.5 Open Science Practices

Open science and open innovation are important for all consortium partners and a key tenet of the MARINER partnership's commitment to accelerating innovation, transparent collaboration, and ensuring the broadest possible impact. The project's core partners (research institutes, universities, and industrial stakeholders) have extensive experience in open-access publishing, open-source software, and data sharing, which will be used to ensure compliance with Horizon Europe's open science requirements. MARINER's open science approach is summarized below.

Early and Open Sharing of Research: Pre-registration will be applied to all major lab testing campaigns (T5.2 and T6.2). The initial step before testing will in all cases be to tailor existing testing protocols to maritime applications, especially for system-level Accelerated Stress Testing (AST) and short-stack trials and register the resulting protocols and results in open repositories (we will use **OSF**²⁴ or **Zenodo** and the **Clean Hydrogen JU Data Hub**). This ensures transparency regarding our methodology prior to obtaining final results. Preprints and publications will be actively encouraged on platforms such as Zenodo, arXiv, or HAL, prior to peer review, to ensure broad dissemination of interim findings. Final peer-reviewed articles will be made openly accessible through gold or green Open Access (OA) routes, in compliance with Horizon Europe mandates. Moreover, thanks to the Couperin consortium²⁵ agreement, MINES, and CEA can publish scientific papers with major publishers without paying Article Processing Charges (APCs). We expect to submit at least two open-access papers from these institutions. Conference papers and workshop presentations will disseminate project findings (including durability assessments, multi-stack control strategies, TCO/LCA analyses) in major events such as the European Hydrogen Week and maritime decarbonization forums.

Open Source Software and tools: VirtualFCS, an open-source platform developed in the eponymous project, will serve as the basis for integrating both data-driven and physics-based fuel cell models, capable of simulating multi-MW maritime fuel cell systems in **WP7**. Source code and manuals will be released under a permissive open-source license (for example, **GPL** or **MIT**) through GitHub, with a digital object identifier (DOI) assigned (via **Zenodo**²⁶) to ensure findability and citation. In addition, algorithms for real-time monitoring, diagnostics, and control will be published as open-source libraries, enabling significantly simplified adoption and continued innovation.

Engaging All Knowledge Actors: MARINER will involve industry stakeholders, including shipping companies, classification societies, and maritime technology providers, through structured workshops (**WP9, WP11, WP12**) and the public release of aggregated results, ensuring iterative feedback loops and paving the way for subsequent



Figure 6: MARINER's existing connections for stakeholder engagement and user feedback.

²⁴ <https://osf.io/>

²⁵ <https://www.elsevier.com/open-access/agreements/france-couperin-consortium>

²⁶ <https://zenodo.org/>

deployment. The scientific and educational community will be engaged through training sessions (T12.2) in advanced simulation and control software, enabling universities, RTOs, and SMEs to replicate and build upon our findings.

1.2.6 Research Data Management Plan (DMP)

MARINER will produce, manage, and share a wide range of datasets and code repositories. To ensure this is done in a consistent manner in accordance with the FAIR (Findable, Accessible, Interoperable, Reusable) principles, consistent with the Horizon Europe guidelines, and with the consortium's own confidentiality and IPR requirements, a DMP will be developed, and later updated. This work will be led by the Project Coordinator (NORCE) in close collaboration with the relevant consortium partners, and significantly, the stakeholder advisory board, to facilitate effective dissemination and interoperability of the data sources beyond the consortium.

The main data types include:

1. Short-stack and submodule measurements (e.g., polarization curves, operating conditions time series, failure modes) for WP3 and WP5. Estimated size: 5 GB
2. Accelerated Stress Testing (AST) data for WP5 (e.g., sensor readings, environmental conditions of the climatic chamber, load profile applied). Estimated size: 50 GB
3. Logs from the 1 MW endurance test from WP6 (real-time power output, efficiency metrics, operational conditions). Estimated size: 50 GB
4. Stack and system level simulation models (the GENEVOS stack specific data driven degradation model), and the 10MW system level simulation model based on VirtualFCS)
5. Control and monitoring software code from WP7.
6. Techno-economic and life cycle assessment data from WP8 (covering cost figures, bills of materials, and emissions).

The DMP evolves over the course of the project, with an initial version delivered at M6, and kept up to date through the project as a living document, updated every year or when project results require it. Publicly shareable data will be given standardized access protocols that distinguish between openly available datasets and restricted materials containing commercial or IP-sensitive information. Each task leader will be responsible for monitoring data quality and consistency relative to the DMP for associated deliverables. Confidential or IP-sensitive content will remain protected under password restrictions and explicit request procedures. The consortium ensures five-year minimum availability of validated final datasets after project completion, and the coordinator (NORCE) and WP leaders maintain oversight of data-handling compliance. MARINER does not involve personal or sensitive human data but remains vigilant about anonymizing any user feedback or interviews that might arise during demonstration-related tasks and in end user involvement. By enforcing this DMP, MARINER guarantees transparent, high-quality data management that respects commercial interests while promoting the replicability of research findings across the maritime and broader energy communities.

#§CON-MET-CM§# #§COM-PL-CP§# #§REL-EVA-RE§#

2. Impact #@IMP-ACT-IA@#

2.1 Project's pathways towards impact

2.1.1 Impact Pathway

The project's results will make a unique contribution towards (1) the outcomes specified in the call topic, and (2) the wider impacts, in the longer term, specified in the respective destination in the work programme.

Project's contribution to the topic outcomes

Outcome OC#1 Development of low-cost, efficient, and flexible multi-stack FCS architectures suitable for multi-MW deployments, aiming for full-scale demonstrators compatible with end-user requirements by 2030;

Project contribution: MARINER will develop a modular fuel cell system with a continuous power rating in the range 100-300 kW, at a CAPEX below 1000 EUR/kW. A 1 MW multi-stack demonstrator will showcase continuous operation at 55% LHV efficiency and with degradation rates indicative of >60,000 h lifetime. Meeting StaSHH fuel cell system standards for heavy-duty applications and aligned with a roadmap towards type-approval, the design will ensure flexibility across vessel types and operational needs and supports scalability up to 10 MW by system replication, minimizing redesign efforts. Optimised stack number and rating will be defined, along with integrated air, H₂, cooling, and power subsystem architectures. Furthermore, innovative electrical architecture and energy management strategies will reduce system complexity and improve efficiency. Cost targets below 1200 €/kW (CAPEX) will be achieved via common stack frames, shared BoP and widely available components. The approach supports rapid deployment of full-scale, end-user-compatible MW-class FC systems by 2030.

Specific KPIs include: **KPI1.2.** Reduced cost of Fuel Cell module with a continuous power rating 100-300 kW: CAPEX < 1000 EUR/kW; **KPI 1.4** - 1 MW continuous operation at 55 % LHV system efficiency; **KPI 1.5** - KPI1.6: 1 MW multi-stack system based on StaSHH standards > 60.000 h of system lifetime at a CAPEX < 1200

EUR/kW.

Outcome OC#2 Further strengthening and consolidating the European fuel cell system supply chain, thereby securing European industry's competitiveness and strategic independence in critical technologies in a global market for large (MW) scale fuel cell systems;

Project contribution: The project leverages scalable architecture to streamline manufacturing and enable cost-efficient multi-megawatt systems. A flexible, modular BoP supports scaling via interchangeable sub-systems, enhancing supply chain adaptability. The use of off-the-shelf components and design reuse lowers CAPEX and boosts domestic component sourcing. Smart thermal and water management supports durability and responsiveness, reducing operational risks. European industry will gain strategic autonomy thanks to type-approved (by 2030), class-certified fuel cell systems for export and deployment. Furthermore, by combining numerical models, advanced diagnostics, innovative control, and a safety-by-design approach, MARINER will establish a new benchmark for the design and operation of large-scale maritime fuel cell systems. In addition, links with relevant EU initiatives (e.g. ZEWT, StaSHH, HyShip, H2MARINE) will maximise knowledge transfer and impact. These outcomes will support European industry leadership, enabling the delivery of reliable, safe, and long-lasting fuel cell technologies that meet the performance, regulatory, and operational demands of the maritime sector. MARINER will hence strengthen EU leadership in critical clean-tech systems, aligned with Fit-for-55 and maritime decarbonisation goals.

Specific KPIs include: **KPI 1.1** - Compliant with SatSHH standards; **KPI 1.3** - Marine compliance and roadmap towards type-approval; **KPI 1.6** - CAPEX < 1200 EUR/kW; **KPI 5.6** - Presentation of project results in at least 5 European working groups (incl. standardisation technical committees)

Outcome OC#3 Providing more robust, durable and lower cost MW scale fuel cell systems suitable for future integration in the 10s of MW scale in maritime applications;

Project contribution : Durability will be demonstrated through real-world operational scenarios and comprehensive system-level Accelerated Stress Testing (AST). AST data will feed into advanced numerical models combining physics-based and data-driven approaches. These tools will accurately predict long-term degradation, identify key durability drivers, and optimise wear scenarios. An optimized power architecture enhances efficiency and durability and allows continued operation despite stack or DC/DC module faults. Flexible, scalable BoP systems enable smooth multi-MW integration while reducing cost and complexity. Smart water and heat management ensure rapid load response without compromising system integrity. The use of standardised modules and off-the-shelf components lowers CAPEX and simplifies multi-megawatt scalability.

Specific KPIs include: **KPI 1.4** - 1 MW continuous operation at 55 % LHV system efficiency; **KPI 1.5** - > 60000 h of system lifetime; **KPI 1.6** - CAPEX < 1200 EUR/kW.

Outcome OC#4 Encouraging demonstrations that lead to broader local, regional, and Union-wide deployment in various transport sectors;

Project contribution: MARINER will advance its fuel cell system from TRL 4 to TRL 6 through a structured demonstration process, including accelerated stress tests at module level and full-scale validation of a 1 MW multi-stack system under realistic maritime conditions. Testing will be conducted across three specialised laboratories, enabling systematic assessment of efficiency, durability, reliability, and safety. To ensure market relevance and support future TRL 7–9 progression, MARINER integrates two operational use cases from SCORPIO and KIARA, representing distinct vessel types and mission profiles. Developed in collaboration with end-users, these use cases will guide design optimisation, regulatory alignment (WP10), and economic and environmental assessments (TCO, LCA in WP8), supporting commercial readiness.

MARINER will also explore spillover potential into other transport sectors, evaluating system adaptations for cross-sector applicability. User and regulatory feedback will be captured via an Advisory Board and stakeholder consultations, supporting the creation of best-practice guides and alignment with industry needs. Finally, a dedicated review of regulatory gaps will inform system design and de-risk deployment. By coupling rigorous technical demonstration with two representative maritime use cases, MARINER directly supports regional and Union-wide ambitions for clean propulsion. The project will provide a validated, scalable FCS architecture ready for replication across a range of vessels and routes, thus laying the groundwork for widespread multi-MW deployments across the EU maritime sector and beyond, aligning with regional and EU decarbonisation objectives in the transport sector. **Specific KPIs include:** 3 testing environments engaged for pretesting, validation and real-world operational demonstration over 2 campaigns and operational use cases, including a full 1 MW system tested under real maritime conditions. Development of Fuel cell system AST protocol for maritime application: **KPI 3.1** - Validation with at least 1000h in relevant environmental conditions (moistly, salty, different air inlet temperature); **KPI 4.3** - Procedures, recommendations and best practices for end users of the 1 MW FCS; **KPI 4.2**

- Comparison of TCO and GHG emissions for at least two segments (offshore and cruise) and two alternatives (biofuels, batteries, ammonia ICE or SOFC).

Outcome OC#5 Facilitating the development of and feeding into European and international regulations, codes, and standards for wide spread use of hydrogen and large scale fuel cell systems;

Project contribution: MARINER will ensure that safety, reliability, and compliance are central to its fuel cell system design and validation approach. Through Lloyd's Register's risk-based design (RBD) methodology and the technical expertise of a system integrator and of a FCS OEM, the project will conduct comprehensive safety assessments, identifying critical failure modes and redundancies for modular multi-MW systems. A review of regulatory frameworks and gap analysis will inform design choices and feed back into system architecture with evolving maritime hydrogen standards. Complementary risk assessments and safety analysis will lead to a consolidated safety analysis report, guiding on-board integration and compliance. To enable broader uptake, MARINER will maintain active dialogue with European Standardisation Organisations (CEN, ISO) technical committees, and disseminate key findings through workshops and stakeholder engagement activities. While direct standardisation will occur post-project, MARINER will create a robust knowledge base and validated best practices to prepare the ground for future regulatory convergence and harmonised certification pathways, supporting Europe's competitiveness in the fuel cell market and the development of the hydrogen value chain.

Specific KPIs include: Number of standardisation/technical committees engaged or informed: ≥ 2 committees (CEN, ISO, etc.); **KPI 4.3** - Procedures, recommendations and best practices for end users of the 1 MW FCS;

Outcome OC#6 Facilitating cross-sector collaboration and knowledge transfer, supporting industry-related skills, and enhancing Small and medium Enterprises' (SME) involvement in the hydrogen economy;

Project contribution: MARINER will actively foster cross-sector collaboration by coordinating with key maritime hydrogen projects (e.g., H2MARINE, StaSHH, HyShip, FLAGSHIPS) to ensure synergy, complementarity, and knowledge sharing. Moreover, members of MARINER are taking part in Horizon Europe ZEWT projects and in the cluster EU Waterborne Transport Synergies Ecosystem (e.g. CEA via NEMOSHIP²⁷), see Section 1.2.2 for a full list. A structured collaboration framework, as well as the involvement of consortium members in some of these projects (e.g. CERTH in H2MARINE, CEA in StaSHH, etc.), will support the exchange of best practices and technical insights, avoiding duplication of effort. To promote skills development, MARINER will deliver hands-on training on fuel cell modelling and control using the VirtualFCS platform, targeting OEMs, integrators, and end users. A dedicated task will ensure knowledge transfer to standardisation bodies and industrial stakeholders, including SMEs, through targeted events and technical materials. The project will engage widely via peer-reviewed publications, conferences, and stakeholder workshops. Special attention will be given to enhancing SME involvement and upskilling industry players, laying the groundwork for broader hydrogen economy integration beyond the maritime sector.

Specific KPIs include: **KPI6.2** - Number of cross-sector partnerships established (e.g., with other EU-funded projects, industry consortia, or clusters): ≥ 5 partnerships; **Number of SMEs engaged** in the project activities (workshops, use case contributions, dissemination, etc.): ≥ 10 SMEs; **KPI5.4** 1 training session for OEMs, SMEs, integrators, ship operators organized with ≥ 10 participants.

Outcome OC#7 Improvements in design, diagnostics and monitoring procedures of FCS (also looking at innovative measuring / sensor devices at this purpose);

Project contribution: MARINER will leverage digitalisation, advanced modelling, and smart control strategies to achieve an FCS lifetime of up to 80,000 hours. Central to this impact is the development and implementation of numerical models of the fuel cell system. These include both physics-based models and data-driven models (e.g. anomaly detection autoencoders), as well as hybrid approaches (e.g. surrogate models). They will simulate fuel cell behaviour under realistic maritime loads and environmental conditions, including degradation mechanisms at both the stack and system level. These models will be calibrated and validated using Accelerated Stress Testing (AST) results at short-stack and full-system scale and will support the extrapolation of long-term performance and durability across different use cases (e.g., ferries, cargo vessels, offshore support vessels). These models will be integrated into the open-source simulation platform VirtualFCS, enabling system designers and integrators to evaluate critical trade-offs between efficiency, durability, cost, and reliability during the early design phase. To ensure robust and safe operation, MARINER will incorporate health-conscious control strategies that dynamically manage load-sharing, thermal balance, and water management across multiple fuel cell modules. The multi-scale energy management system (EMS) will support independent control and reconfiguration of modules in the event of partial faults, supporting redundancy and fault tolerance — essential for critical maritime operations. Safety and reliability are also embedded into the system from the design stage. A comprehensive failure mode and effects

²⁷ <https://nemoship.eu/euwtsynergiesecosystem/>

analysis (FMEA) and reliability analysis will be conducted, supported by a review of classification requirements and regulatory gaps. These insights will guide the inclusion of built-in safety mechanisms, robust control logic, and monitoring sensors, ensuring fail-safe system behavior in compliance with maritime standards. Tests will be done under operationally relevant maritime scenarios, enabling direct validation of system response, monitoring precision, and control robustness. The system's diagnostic architecture will incorporate innovative sensors and real-time monitoring algorithms that support predictive maintenance, lifetime estimation, and early warning of failure modes, significantly reducing OPEX and downtime. By combining numerical models, advanced diagnostics, innovative control, and a safety-by-design approach, MARINER will establish a new benchmark for the design and operation of large-scale maritime fuel cell systems. These outcomes will support European industry leadership, enabling the delivery of reliable, safe, and long-lasting fuel cell technologies that meet the performance, regulatory, and operational demands of the maritime sector.

Specific KPIs include: **KPI2.1** - RMSE < 10 mV for beginning-of-life polarization curve and load-cycle simulation; **KPI2.2** - RMSE < 20 mV for EOL polarization curve and load-cycle simulation; **KPI2.3** - Load-cycle simulation > 100x faster than real-time; **KPI2.5** - Lifetime at system level: 80.000 hours.

Outcome OC#8 Improvements of testing protocols for the quantification of FCS performance and lifetime in maritime environments, including accelerated stress tests;

Project contribution: Durability in real-time conditions will be evaluated by defining and using System-Level Accelerated Stress Test (SL-AST) protocols that reflect the real operational conditions faced by ships/maritime application—including humid, air inlet temperature variations and conditioning with salt, and dynamic load cycles. These protocols will run in at least 1,000h and with FC system degradation accelerated by a factor of 20. Data-driven models fed with AST results will extrapolate degradation of a system lifetime to over 40,000 hours. Unlike conventional AST focused on materials, MARINER's SL-AST methodology integrates BoP-stack interactions and multi-stressor environments to capture system-level failure modes. The developed model will also support design optimisation, predictive maintenance, and health-aware control strategies. The protocols used, test results and lessons learned could lay the groundwork for future harmonised AST methods for maritime applications that could be integrated into industry standards.

Specific KPIs include: MARINER introduces a non-intrusive, system-level AST methodology that goes beyond material-focused lab tests by incorporating BoP-stack interactions and multi-parameter stress conditions—filling a critical gap in standard practices. SL-AST protocols are tailored to real-world maritime conditions, specific KPIs include: **KPI3.1** - Validation with at least 1000h in relevant environmental conditions (moistly, salty, different air inlet temperature); **KPI3.2** - > 20x acceleration of fuel cell system degradation; **KPI3.3** - Extrapolation of degradation for more than 40000 hours; **KPI3.4** - Fuel cell system lifetime estimation for different use cases.

Outcome OC#9 Improvement of overall system performance of FCS in order to increase the availability and durability and meet the needs of naval and maritime end users.

Project contribution: MARINER will significantly improve the performance, availability, and durability of fuel cell systems for maritime use by developing a modular, scalable architecture capable of continuous operation at ≥55% LHV efficiency and targeting lifetimes of up to 80,000 hours. The system integrates smart diagnostics, health-conscious control, and predictive maintenance features to reduce downtime and ensure high availability, even in the event of partial faults. Through accelerated stress testing and real-world validation under maritime conditions, MARINER methodology will capture system-level failure modes, aiding the development of digital twins, predictive maintenance, and health-aware control, ensuring robustness, reliability, and operational relevance. Demonstrations will be guided by two operational use cases (short-sea and cruise vessels), ensuring diverse application relevance. This facilitates scalability toward 5–10 MW systems and beyond. The validation results, aligned with regulatory pathways and techno-economic analyses, will reduce uncertainty for shipowners and integrators, laying the groundwork for uptake in both regional and Union-wide maritime decarbonisation strategies. These innovations will enable the deployment of fuel cell systems that meet the stringent requirements of both naval and commercial maritime end users.

Specific KPIs include: **KPI 1.3** - Marine compliance and roadmap towards type-approval; **KPI 1.4** - 1 MW continuous operation at 55 % LHV system efficiency; **KPI2.5** - Lifetime at system level: 80.000 hours.

Project's contribution to impacts expected in the destination

The MARINER project will contribute directly to the objectives of the European Green Deal and the Fit for 55 package, by enabling the deployment of scalable, low-emission fuel cell technologies in the maritime sector. It aligns with the EU Hydrogen Strategy and supports the goals of ReFuelEU Maritime and the Sustainable and Smart Mobility Strategy by demonstrating reliable, durable, and economically viable 1 MW-scale PEMFC systems for maritime applications.

2.1.2 Scale and significance

The MARINER project directly addresses critical barriers to the large-scale deployment of fuel cell systems in the maritime sector, delivering innovations in system architecture, cost-efficiency, durability, diagnostics, and regulatory alignment that together represent a transformative step for European hydrogen technologies. Its outcomes align with urgent decarbonisation mandates (FuelEU Maritime: -80% GHG intensity by 2050; IMO: net-zero by 2050), while positioning European industry at the forefront of a fast-growing global market.

Market Demand: The global merchant fleet (>100 GT) includes over 126,000 vessels, of which 84% are small to medium-sized. Around 40% are >25 years old, with 60% of those in the small vessel category, implying substantial near-term retrofit and new-build opportunities. MARINER's scalable system targets this segment, where an estimated 5% market penetration by 2035 would yield >5,000 vessels suitable for fuel cell integration.

Strategic European Industrial Autonomy: MARINER will demonstrate a 1 MW multi-stack PEM fuel cell system at 55% LHV efficiency, with a >60,000-hour lifetime, and CAPEX targets below €1000/kW — representing a significant performance leap and cost breakthrough. Its modular design supports scalability to 10 MW through replication, enabling cost-effective adaptation across diverse vessel types and mission profiles. These figures are aligned with the Clean Hydrogen Partnership KPIs and pave the way for full-scale commercial deployments by 2030. Thus, MARINER will be able to reinforce Europe's competitiveness in a global market projected to reach €2.5–3 billion by 2032 for marine fuel cells. By adopting standardised components, shared BoP, and engaging SMEs and supply chain actors, the project will support domestic manufacturing capacity, strategic independence in critical technologies, and the development of class-certified systems for export. Furthermore, through close engagement with classification societies (e.g., Lloyd's Register) and standardisation bodies (CEN, ISO), MARINER will deliver best practices, a Safety Analysis Report, and a new AST methodology for maritime FCs, filling regulatory gaps and accelerating type approval and certification. This de-risks market entry and supports EU leadership in global rule-setting for hydrogen-powered ships. The integration of digital modelling, predictive diagnostics, and open-source tools (e.g. VirtualFCS) will also ensure broad usability and long-term innovation impact.

Environmental and Societal Impact: LCA comparison for an offshore vessel using green hydrogen compared to a diesel equivalent shows that such hydrogen power module can save 85% CO₂ footprint, 84% less ionising radiation, and 96% reduction in key air pollutants over a 20,000-hour operational lifetime (3600 hours per year), supporting broader health, climate, and port decarbonisation goals.

2.1.3 Barriers and requirements to reach impact

Factor	Requirements for impact achievement	Potential barriers for impact achievement	Mitigation measures to limit the impact of potential barriers
Political	Government support for zero-emission maritime technologies and funding of hydrogen infrastructure	Delays in implementing maritime climate regulations across EU member states	Early- engagement with EU policymakers and national maritime authorities thanks to the AB. Use case scenarios adapted to different national context.
Economical	Competitive Levelised Cost of Energy (LCOE) and Total Cost of Ownership (TCO) compared to diesel engines and LNG.	High CAPEX of FCS and clean H ₂ , underdeveloped H ₂ value chain.	TCO analysis to demonstrate long-term value. Engagement with green shipping finance initiatives for the scalability roadmap
Social	Training and upskilling for SMEs in the maritime sector, crew and technician in FC technologies and control. Need for inclusive capacity building.	Workforce resistance to change, Limited public awareness of hydrogen safety and benefits	Organisation of a 2-days training, elaboration of a pedagogical toolkit and best practices guide, dissemination to professionals network.
Technological	Robust modular FC architecture with fault tolerance and plug-and-play capabilities. Durable operation in harsh maritime environments (humidity, salinity, vibrations).	Limited operational experience at multi-MW scale in real ship environments. BoP complexity and degradation under dynamic load cycles. Interoperability with existing shipboard systems (electrical,	Multi-lab and full scale testing under real maritime conditions will be conducted, together with durability and endurance analysis. At least 2 use case scenarios will be assessed. The modular design will facilitate scaling, maintenance and disassembly.

		thermal, automation).	
Ethical	Data transparency, especially related to digital twins, AI, and performance monitoring. Responsible use of rare materials	Potential for regional economic inequality if hydrogen infrastructure is unevenly developed. Environmental justice concerns if fuel supply chains displace pollution or resource extraction abroad.	Publication of LCA results, inclusion of environmental and social dimension in exploitation activities. Engagement with ports authorities.
Regulatory	Type approval for modular FC systems. Safety and certification protocols adapted to multi-stack configurations.	Regulatory lag in certifying new tech architectures and fuels.	The participation of a classification society in the project ensures that MARINER results will feed prenormative activities.

2.2 Measures to maximise impact - Dissemination, exploitation and communication #@COM-DIS-VIS-CDV@#

Measures to maximise the project impact will cut across all WPs, and especially WP9, 11 and 12. MARINER will submit a full Dissemination and Exploitation Plan (PDE), including the communication activities at M6.

2.2.1 Target groups

Partners have identified six Target Groups (TG) for dissemination activities, and have pre-designed a plan on how to reach them, what messages to send them, and what key take away/ benefits they will gain from engaging in/following the project activities.

TG1: Equipment manufacturers and system integrators

OEM companies (such as GENEVOS) that design and manufacture the core technology, such as fuel cell stacks, power electronics, control systems, etc. as well as companies responsible for assembling and adapting fuel cell technologies into full, operational shipboard systems (such as VARD). **Access to TG:** The presence of GENEVOS and VARD in the consortium ensure access to fuel cell component supply chains, technical decision makers, and other system integrators thanks to their involvement in associations such as ZESTAs the Zero Emissions Ship Technology Association or the Fuel Cell and Hydrogen Energy Association. **Key interest:** low-cost, efficient, and flexible multi-stack FCS architectures suitable to multi-MW deployment, roadmap to TRL 9 and type approval. **Key messages:** MARINER offers a validated route to multi-MW PEMFC integration in vessels as well as procedures and best practices. The project addresses modularity, redundancy, and maritime safety challenges and supports safe and scalable system integration for multiple ship types. **Key benefits:** Recognition in maritime hydrogen applications, Early-stage feedback to support next-gen development.

TG2: Other actors of the maritime industry: from shipbuilders to seaports

Shipowners and operators, shipbuilders and marine engineering firms, ports and terminal operators. **Access to TG:** Through the participation of GENEVOS, VARD - part of the Fincantieri Group, MARINER integrates a certified OEM, a top-tier system integrator and shipbuilder with global reach. Their embedded industrial networks (e.g. GICAN the French shipbuilding industry association, ZESTAs, etc.), combined with direct contributions from partners involved in H2MARINE and HySHIP projects (among others), ensure privileged access to a broad community of shipowners, designers, and commercial and technical operators. This access is operationalised through planned use-case demonstrations with Scorpio Group and KIARA, as well as structured stakeholder engagement via our Advisory Board. **Key interest:** Primary beneficiaries and implementers of the MARINER technology to comply with the FuelEU maritime regulation and reach the 2030 IMO targets to reduce GHG Emissions from Ships. **Key messages:** MARINER helps accelerate compliance with marine hydrogen regulations. Delivery of reliable, safe, and long-lasting fuel cell technologies. Emphasis on the lower price and on a modular design ensuring flexibility across vessel types and operational needs. **Key benefits:** Potential participation in a follow-up project as a use case owner.

TG3: Scientific and research community

Academia, RTO, and R&D teams from other industries. **Access to TG:** All partners have extensive contacts with the European R&D&I community specialised in energy systems, hydrogen, electrical engineering, manufacturing and maritime applications. Clustering activities will be organised with StaSHH (CEA), HyShip (NORCE), H2MARINE (CERth), FLAGSHIPS, MARANDA, ShipFC (GENEVOS) and other FC projects thanks to partners' involvement or existing connection in those projects. **Key interest:** State of the art, project results, business development. **Key messages:** Clustering activities and collaboration agreement with at least 5 FC projects. Dissemination of the project advancement and results through the project website, peer-reviewed articles, conferences and the final event. **Key benefits:** orienting existing or future research.

TG4: Policy makers, regulators and standardisation bodies

Maritime safety authorities such as EMSA, IMO Maritime Safety Committee, International Association of Classification societies, IEC Technical committees 105 on Fuel cell technologies, IEC TC 18 on Electrical installation of ships, CEN-CENELEC Coordination Group on Hydrogen, CEN-CLC/JTC 6 – Hydrogen in Energy Systems and TC268 – Hydrogen Technologies, etc. **Access to the TG:** GENEVOS, VARD and Lloyd's Register participate in numerous working groups and technical committees. **Key interest:** Influencing and aligning with evolving policy and standards; Ensure that the 1 MW PEMFC system and its modular arrangement can be installed on real vessels in compliance with marine safety rules. **Key messages:** innovations will be presented to ensure their alignment with existing regulation and standards and ensure their market readiness. **Key benefits:** Enable regulatory compliance as well as future standards and regulation updates.

TG5: Hydrogen ecosystem actors

Hydrogen valleys and initiatives, energy suppliers, hydrogen producers and infrastructure developers. **Access to the TG:** Through project partners involvement in national and European working groups and platforms (e.g. Hydrogen Europe Research; France Hydrogen; EERA JP Fuel Cells and Hydrogen, European Clean Hydrogen Alliance), collaboration with other R&I projects, participation in industry events and the European Hydrogen Week, articles on digital media. **Key interest:** Access to real-world performance data; demonstration results; design integration best practices; certification and regulatory pathways; Understand safety, reliability and modular integration approaches. **Key messages:** MARINER delivers scalable, certifiable 1-MW PEMFC systems tested under maritime conditions with validated SL-AST protocols and digital twin tools. Users feedback are considered in the design stage of the project and best practices guide are developed. **Key benefits:** Scaling up interest in waterborne transport and cooperation between actors of the hydrogen and fuel cell supply chains.

TG6: SMEs and Start-ups

Component manufacturers (BoP, sensors, stacks, etc.), Software firm for monitoring control and simulation, maritime-focused cleantech SMEs. **Access to the TG:** Fuel Cell and Hydrogen Observatory, industry fairs, digital media, publication in technical magazines, link with clusters. **Key interest:** Encourage technology uptake and contribute to supply chain and innovation. **Key messages:** MARINER offers a training programme on software use and pedagogical toolkit as well as procedures, recommendations, and best practices. **Key benefits:** in-depth guidance on optimal software use, ensuring that users can integrate these tools into their operations for maximum efficiency.

TG7: Maritime associations and civil society

Maritime associations such as the European Community Shipowner's Association, the European SeaPorts organisation, GICAN, ZESTASs, and Waterborne; NGOs & initiatives that promote sustainability in transport (Clean Shipping Coalition), international companies in shipping, maritime administrations and maritime routes from MARINER use cases. **Access to TG:** Consortium partners (CEA, GENOVOS, VARD, MINES etc.) are already involved in such networks and associations. When required, the Advisory Board will help the consortium to get in touch with additional organisations. The European research and innovation platform for waterborne industries will also be engaged. **Key interest:** Reach the net zero maritime transport target. **Key messages:** Raise awareness and build acceptance on hydrogen technology through digital communications, involve them in relevant events to disseminate project activities and get their feedback.

2.2.2 Initial plan for communication and dissemination

Several communication tools (mailing list, document sharing platform etc.) will be used to ensure a smooth internal communication between the consortium partners: this will be part of the project coordination. External communication tools (also used for dissemination purposes) will allow to raise awareness among the TGs stakeholders and support the dissemination activities (see below).

COMMUNICATION MATERIALS	
Project documentation	All stakeholders
All public project deliverables will be made available on the project's website. Public summaries of each deliverable will be provided in the website's news and advertised on social medias to encourage stakeholders to engage in our activities.	
Flyer	TG1, TG2, TG5, TG6
A flyer will be developed to present the project's main activities in a visual way: it will be distributed in small amounts during in person fairs and events. The online version will be used when contacting stakeholders directly per email, to facilitate their project's comprehension.	
Final brochure	Industrial actors, SMEs, R&D&I community (TG1 to TG6)
A final brochure will be created in M47 based on the best practice guides, procedures and recommendations. It will present all project results and be distributed during the final event, but also sent to relevant stakeholders (members of the Advisory Board, targeted policy makers etc.)	
COMMUNICATION DIGITAL CHANNELS	
Project's website	All stakeholders
The project's website will be the main dissemination channel: it will be linked to social medias and display information about other dissemination tools and activities, such as links to register to events, access the MOOC, register to and read the newsletter, access publications etc. Associated KPI: 5000 visits during the project, monitored through a dedicated tool such as Matomo (more GDPR compliant than Google Analytics)	
Social medias	All stakeholders
The project will primarily use LinkedIn (to exchange with professionals/experts), Youtube (to exchange with the general public) as social medias (VARD, Mines Paris YouTube Channels will mostly be privileged). These will disseminate general information on the project and refer to news on the website/ events (inviting stakeholders to register)/ public deliverables/ videos etc. These snapshots will target both stakeholders that the project partners do not know, and stakeholders that are already aware on the project to keep them updated and interested. Associated KPI: 300 followers, monitored through the social media platforms	
Newsletters	All stakeholders
4 newsletters will be developed during the project (one every year + one at the end of the project), to remind the stakeholders interested in the project of the latest project's activities and achievements. These newsletter will be available on the project's website and disseminated through the project's and partners social medias.	
Zenodo	R&D&I (TG1, TG2, TG3) and policy makers
All public deliverables and publications of the project will be uploaded on a dedicated Zenodo group to ensure their long-lasting findability.	

MARINER dissemination activities will have three major aims: ensure the project's alignment with environmental, economic and social concerns, prepare the ground for the project's scale up and maximise the impact of the research conducted in the project, to facilitate its replicability.

DISSEMINATION ACTIVITIES	
Webinars	All stakeholders
3 webinars will be organised targeting various groups of stakeholders to introduce the methodologies, share good practices, to network and plan common activities etc. Associated KPI: at least 3 webinars organised.	
Workshops/Clustering Activities	TG1, TG2, TG4, TG7
At least 4 workshops will be organised to gather user input and feedback at regulatory and non-technical level, involving advisory board members, relevant projects such as H2MARINE, ZEWT, StaSHH, HyShip, as well as standardisation committees to ensure knowledge exchange and build together the roadmap to higher TRL. Associated KPIs: Cooperation agreements signed with at least 5 projects; 4 stakeholder workshops organised.	
Publications	R&D&I (TG1, TG2, TG3)

6 scientific publications will be produced during the project, mainly linked to WPs 3, 5-10. These publications will be published either in scientific journals or in peer-reviewed platforms, such as the Open Research Europe Platform (see section 1.2.5 for Open Science practices). These publications will mostly address the scientific community, showing progress in the state-of-the-art. Interested journals could be Applied Energy, Journal of Power Sources, Energy Conversion and Management, International Journal of Electrical Power & Energy Systems, International Journal of Hydrogen Energy, Energy Reports, Controls, IEEE Sefet, Sustainable Energy and Powertrains. **Associated KPI:** 6 publications, to be found on peer-reviewed platforms with links gathered on the project's website

Training workshops	OEM, integrators and ship managers
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Training workshops will be organised in MINES Paris premises or at use case sites to train OEMs, integrators, ship managers and other end-users on fuel cell modelling and control. Parts of the training will be filmed, or made into instructional videos, which will be made available publicly on Youtube. A pedagogical toolkit will accompany the training. The promotion will be done through the European Hydrogen Observatory.

Associated KPI: at least 10 participants trained.

Best practices guide	Maritime industry
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Procedures, recommendations and best practices will be packaged in a practical format targeting actors of the maritime industry. Recommended practices, Do's and Don'ts, Illustrative examples from the use cases as well as open-source tools developed will be introduced. **Associated KPI:** best practices and recommendations disseminated towards 50 organisations.

Participation in events	R&D&I community, maritime industry
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The project partners will participate in at least 10 professional fairs and events to network with interested stakeholders and present the project. Interested events could be: Sustainable Energy and Powertrains conference, IEEE International Conference on Sustainable Energy and Future Electric Transportation European Hydrogen week, Hydrogen technology expo, Hyvolution, Connecting Hydrogen Europe, European Fuel Cells and H2 conference, EFCF conference, European Shipping Summit, SMM Trade Fair, Sea Tech Week, Monaco HA, Elect&Hybrid, MEET, GICAN, etc.

Associated KPI: participation in at least 10 fairs/events.

Final event	OEM and integrators, R&D&I community, policy makers
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A final event will take place at the end of the project, to present the project's results and stimulate the project's exploitation and replication. It will be co-located with a larger event.

Associated KPI: at least 90 participants, monitored through registration.

2.2.3 Initial exploitation plan

The project team has already identified the **13 main specific results** that will be produced during the project. For each of these results, a possible exploitation path has been identified, along with the associated exploitation targets:

WP	Result	(co-) Owners	Potential end-users/markets	Foreseen exploitation routes
3, 4	R#1 Standardized fuel cell module configuration for low cost and extended lifetime – 200 kW prototype and assembly documentation	GENEV OS	OEMs, Systemintegrators, Ship owners and ship Builders, Ship architects and consultancies specialised in retrofitting	<i>At the project end:</i> · Upgrade of GENEVOS existing high-power module with a more scalable and lower cost system · Route: Commercial sale · IP protection: Patent-protected concept – refer to section 1
3, 4	R#2 Scalable and modular multi-stack system for multi-MW maritime applications – 1 MW demonstration system	GENEV OS, VARD		

4	R#3 Class-reviewed FAT protocol	CEA, LR EMEA	OEMs, System integrators, Classification and Certification Societies, RD&I community	At the project end: - Dissemination through open-access publication. - Use in further research.
5	R#4 Extended Accelerating Stress Testing protocol adapted to FCS in maritime application	MINES, CEA	OEMs, Marine Engineering Services, RD&I community and testing facilities	At the project end: - Dissemination through open-access publication - Liaison with JRC for the development of harmonised test protocols for maritime applications.
6, 7	R#5 OS Multi-MW FCS performance and degradation physics-based models	USTUT T, MINES, CEA, SEC	OEMs, RD&I community, testing facilities,	At the project end: - Dissemination through open-source publishing (with user manual and API) and inclusion in the VirtualFCS library. - Development and organisation of a 2-days training, with its pedagogical toolkit translated in consortium national languages.
6	R#6 Data-driven/hybrid FCS degradation models (FCS digital twin)	CEA, USTUT T, MINES	OEMs, RD&I community, testing facilities,	At the project end: - Dissemination of methodology through open-access publications. - Exploitation by GENEVOS and VARD in the FCS design process (feedback loop).
6	R#7 Test plan for demonstration at lab facility, under simulated marine conditions	VARD, CEA, SEC, MINES	RD&I community, Testing facilities, Marine Engineering Services	By the end of the project: - Liaison with JRC for the development of harmonized test protocols for maritime applications. - Use for further research
7	R#8 Open-source multi-level fuel cell SoH monitoring framework and control algorithms	MINES, CEA, USTUT T	OEMs, System integrators, Marine Engineering Services, RD&I community	At the end of the project: - Open-source publishing, with user manual and API. +1 year after the project end: - VARD will upgrade its existing software, enabling it to provide a User Interface.
3,5, 6,7	R#9 Datasets of durability and endurance tests	CEA, Mines, SEC, Genevos, LR EMEA	RD&I community, testing facilities, OEMs, system integrators	By the end of the project - Internal use to refine the design - Internal use to study the outline expansions and scalability - Open Access publication of non-confidential datasets
8	R#10 TCO and LCA protocols	CERTH	RD&I community	At the end of the project - Open-access publication of the methodology - Use for further research
9	R#11 Scalability / TRL 9 Roadmap	NORCE, SCORPIO, KIARA, GENEVOS, VARD	ALL TGs	+1 year after the project end: - Exploitation by GENEVOS for type-approved FCS - Follow-up project for demonstration in vessels

9	R#12 Protocol, recommendations and best practices guide	ALL	OEMs, System integrators, Ship builders & ship owners, Architects specialised in energy retrofit of ships, Policy Makers, RD&I community	At the end of the project: • Publication and dissemination to targeted stakeholders including policy makers to guide future regulation.
10	R#13 Safety analysis and recommendations	LR EMEA	Maritime Community using FCS systems, Classification societies	At the end of the project: • Improve class procedures for assessment of Fuel Power Systems and their modular arrangement on board

The MARINER consortium has a clear strategy to ensure the results of the project translate into long-term industrial and societal impact. Exploitation activities will be embedded throughout the project, coordinated via WP11 and WP12, and regularly reviewed with input from stakeholders and the Advisory Board (WP9). The project aims to support TRL maturation beyond the project timeframe—towards TRL 9—by offering industrial partners (GENEVOS, VARD) and end-users (SCORPIO and KIARA) a unique competitive edge through early access to validated technologies and integration pathways. To maximise market potential, emphasis will be placed on intellectual property protection—via patent filings, trade secret management, and IP monitoring, specially concerning GENEVOS and VARD proprietary design choices at the FCS module scale and electrical integration, respectively. Open-access dissemination will be managed carefully to align with commercialisation objectives.

Business development will be guided by a triple-layered business canvas (economic, environmental, and social), tailored for each exploitable result. A first version has already been developed for the “scalable and modular multi-stack system for multi-

MW maritime applications,” to be commercialised by GENEVOS, with further canvases developed during the project. For the sake of place in the proposal, only the economic business canvas is presented on the right.

The project upscale will build on 4 main outcomes: 1. **A techno-economic validation** through AST, endurance testing, and digital twin models, alongside LCA and TCO analysis,

allowing GENEVOS and VARD to refine the system for real-world integration; 2. **A regulatory and safety analysis**, preparing the ground for future type approval; 3. **End-user scenarios and scalability roadmap** - including the funding strategy - co-developed with ship operators (Scorpio, KIARA), AB members and other FC projects, ensuring relevance and support the scale-up pathway towards full-scale deployment; 4. **Engagement and dissemination of best practices** towards maritime stakeholders (including SMEs), policy makers and standardisation bodies facilitating the uptake in practice.

Research partners (NORCE, MINES, CEA, CERTH, USUTT) will further exploit the results via scientific dissemination and follow-on R&D. SEC and LR EMEA may integrate developed methods into updated safety and certification protocols. Both GENEVOS and VARD foresee the opportunity to upgrade existing product lines, while ship operators have expressed intent to retrofit or build a vessel based on MARINER’s architecture, should technical and economic viability be confirmed. Finally, several actions to be conducted during and after the project to maximise

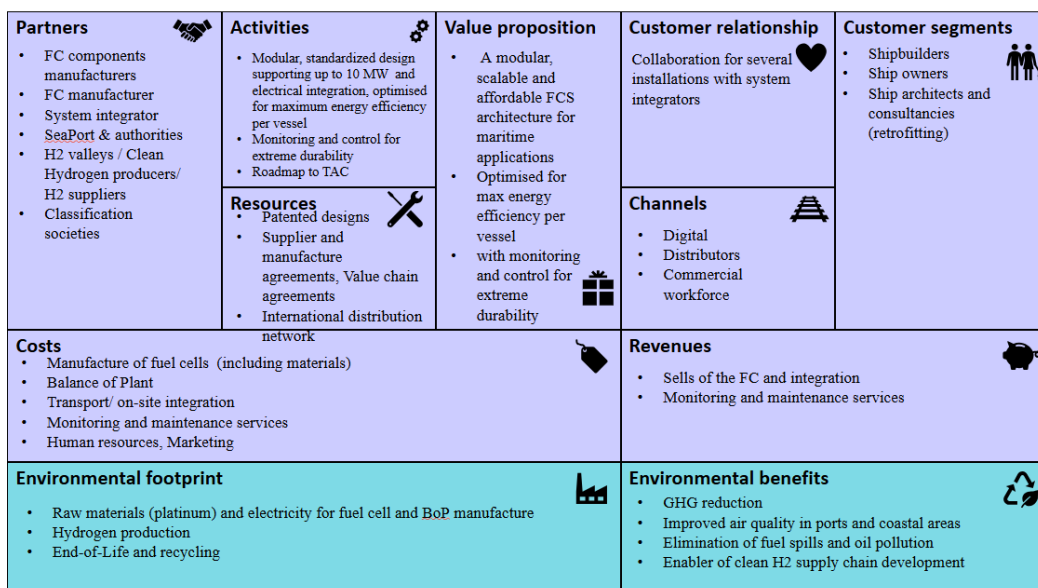


Figure 7: Economic business canvas of the scalable and modular multi-stack system for multi-MW maritime applications

its exploitation have been listed. These actions will be strictly monitored at each of the consortium meetings:

Timeline	Actions to maximise the exploitation
Technical actions	
During the project	Ensure the modularity of the designs for compliance with a large number of vessel types; Develop use case scenarios for future demonstrations in operational environment; Decommission the 1 MW fuel cell system for future demonstration; Risk assessment and safety review; Ensure knowledge exchange with other FC projects, standardisation and other bodies;
After the project	Scale-up a first prototype of the modular multi-stack system for multi-MW maritime applications (TRL9 in 2033); Expand the range of designs to fit other configurations; Demonstration project in at least 2 configurations
Market actions	
During the project	Define the scalability roadmap for class-type approval, outline expansion and further funding; Train the workforce on FCS degradation modelling and control strategies; Develop safety procedures, recommendations and best practices for end-users; Engage actively with policy makers, standardisation organisation, the Advisory Board members and other interested OEMs, system integrators, and end-users; Engage with civil society environmental associations to improve the social acceptance of project results; Participate in conferences and business fairs to present the project and gain interest.
After the project	Follow up with classification societies for class-type approval; Apply to an Innovation Action or Innovation Fund if the results of the project are successful; Engage with private investors and create partnerships with system integrators, seaport and hydrogen valleys.

2.2.4 Strategy for the management of intellectual property

MARINER's IP management strategy will facilitate the identification and protection of exploitable results to yield a positive socio-economic impact. Each consortium partner will manage its own IP using its own process, with DOWEL's support tailored to each partner's needs and capacities. To elaborate the final Result Ownership List and the related IP strategy, consortium-level workshops will be combined with one-on-one support to maximise each partner's ability to exploit project results. Consortium partners will seek to protect and exploit their respective results to the greatest extent possible, while the Consortium and Grant Agreements will enumerate the IP terms, including obligations (e.g., confidentiality, exploitation), ownership, and access rights to background and foreground IP. A preliminary assessment of existing patents has shown that partners have the freedom to perform their tasks according to the action. Early in the project, DOWEL will prepare an IPR Strategy Workshop to help partners identify results, decide whether to exploit or disseminate, and protect results destined for commercialisation. Commercial exploitation and confidentiality will be balanced against open access principles, policy/guidelines, and economic realities (e.g. commercial partners or financing sources). Partners will seek to develop exploitation strategies that accommodate fellow MARINER's partners to the greatest extent possible. GNEVEOS patented design will still apply while for some other results, open-source licensing might be privileged (see below). As much as possible, results will be broken down into units that are owned by one single partner to avoid IP issues. In case joint ownership cannot be avoided, a detailed analysis of the commercial potential of the results, and ambitions of the respective owners will be conducted. As MARINER's nears closure, partners may engage with appropriate external actors (e.g. via license, TTA, spinout etc.) and plan to utilise EU-managed matchmaking facilities to identify end-users and commercialisation partners as needed. DOWEL will assist partners in identifying results for the results ownership list to be submitted to the EC portal at project close.

Open-source licenses handling: The consortium commits to open-source policies, as follows: 1) solutions will be copyright protected using a licensing scheme that will not violate the terms and conditions of the discrete components comprising it; 2) components that can be provided open-source will be delivered. The respective knowledge producer holds the rights to the knowledge they have produced while sharing the software implementation with the rest of the consortium. The consortium will examine the software licenses of all algorithms and components used and decide under which license (or combination of licenses) the solutions will be released. For new components, the "copyleft" license EUPL (European Union Public Licence) seems to be the prominent candidate since it is compatible, business-friendly, and governed by EU law.

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2.3 Summary

SPECIFIC NEEDS	EXPECTED RESULTS	D&E&C MEASURES
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- The maritime sector urgently needs clean, high-power solutions to meet strict GHG targets (FuelEU: -80% by 2050; IMO: net-zero by 2050). - PEM fuel cells show promise but face challenges in cost, durability, and scalability. - Modular designs are key to reduce complexity, ease integration, and support flexible scaling. - OEMs must deliver robust, efficient stack and BoP components at lower cost; integrators ensure compliance and shipboard integration; and shipbuilders or retrofiters must adapt vessels. - Close collaboration is vital to enable market uptake and drive zero-emission shipping.	Validated prototypes: - A modular fuel cell system with a continuous power rating in the range 100-300 kW, at a CAPEX below 1000 EUR/kW. - A 1 MW multi-stack demonstrator showcasing continuous operation at 55% LHV efficiency and with degradation rates indicative of >60,000 h lifetime. - Extended Accelerating Stress Testing and endurance protocol adapted to FCS in maritime application. Improved knowledge: - Harmonised FCS test protocols for maritime application. - Multi-level fuel cell SoH monitoring framework and control algorithms. - TCO and LCA. - Safety recommendations and best practices.	Upgrade of products and services 'Fast track' to commercialisation of the Low-Cost, Modular, and Ultra-Durable Multi-MW Fuel Cell System - Upgraded services (GENEVOS/ VARD/LR/KIARA/SCORPIO) Scalability roadmap Outline expansion, class-type approval, end-use scenarios, spillover potential funding Dissemination towards the maritime industry, actors of the H2 ecosystem, SMEs, scientific communities and regulatory bodies: - Through the project website and social medias - Through dedicated dissemination reports, easy to read for SMEs and policy makers - Through scientific conferences and journals - Through industrial fairs and conferences and training - Knowledge exchange with FC projects, JRC and standardisation bodies, regulator and national authorities
TARGET GROUPS	OUTCOMES	IMPACTS
- Equipment manufacturers and system integrators - Shipowners, shipbuilders, architects and consultancies in ships retrofitting, seaports - RD&I community - Policy makers, regulators and standardisation bodies - Hydrogen ecosystem actors - SMEs and start-ups	- By year 2030, emergence of standard FCS solutions for 10 MW installation and reduced CAPEX (<1000€/kW). - Improved standards, classification procedure and regulation readiness thanks to MARINER insights from testing, safety assessments, and system integration contribute. - A skilled and engaged maritime industry ecosystem to integrate hydrogen technologies into their operations.	Economic: New market for European OEMs, strengthening the H2 value chain, cost reduction Socio-environmental: Accelerated decarbonisation of the maritime sector, aligning with the IMO and FuelEU maritime regulations. Improved air quality in port areas Scientific: Advanced SoA in fuel cell system design, durability modelling and system-level accelerated stress testing. New knowledge in multi-stack control. Open-source simulation tools, methodologies and data to further support the decarbonisation of maritime applications.

#§IMP-ACT-I

3. Quality and efficiency of the implementation # @QUA-LIT-QL@# # @WRK-PLA-WP@#

3.1 Work plan and resources

3.1.1 Overall Structure of the Work Plan

The project is structured into twelve Work Packages (WPs) that cover all necessary steps from project coordination and design, through system build and testing, to commercial, regulatory, and dissemination objectives. WPs 1 and 2, led by NORCE, handle overall project management and coordination (Part A & Part B), ensuring effective oversight, risk management, and communication among the consortium. WP3 (GENEVOS) establishes the system architecture, reliability criteria, and multi-stack design strategies, which then feed into WP4 (GENEVOS) for the FCS build and integration. WP5 (CEA) covers the first part of the test and demonstration campaign and includes accelerated stress testing on a 200 kW subsystem to assess long-term durability at CEA's testing Lab in Nantes. The second part of the

test and demonstration campaign is conducted in WP6 (SEC) which runs a 1 MW demonstration for at least 1,000 hours in a relevant maritime environment at the SEC's Energy House.

Software tasks (WP7, led by USTUTT) uses inputs from the earlier WPs, and builds on existing software developed in other projects, to deliver the control, monitoring, and simulation software for the AST protocol and demonstrations. WP8 (CERTH) carries out techno-economic and life cycle analyses, while WP9 (NORCE) focuses on scale-up feasibility and end-user engagement. WP10 (LR EMEA) addresses safety and regulatory pathways to ensure compliance with maritime standards. Finally, WPs 11 and 12 (DOWEL) handle communication, dissemination, and knowledge transfer, to support market uptake of the developed innovations, and ensuring broad awareness and sharing of the project's test results and datasets while promoting impactful project outcomes.

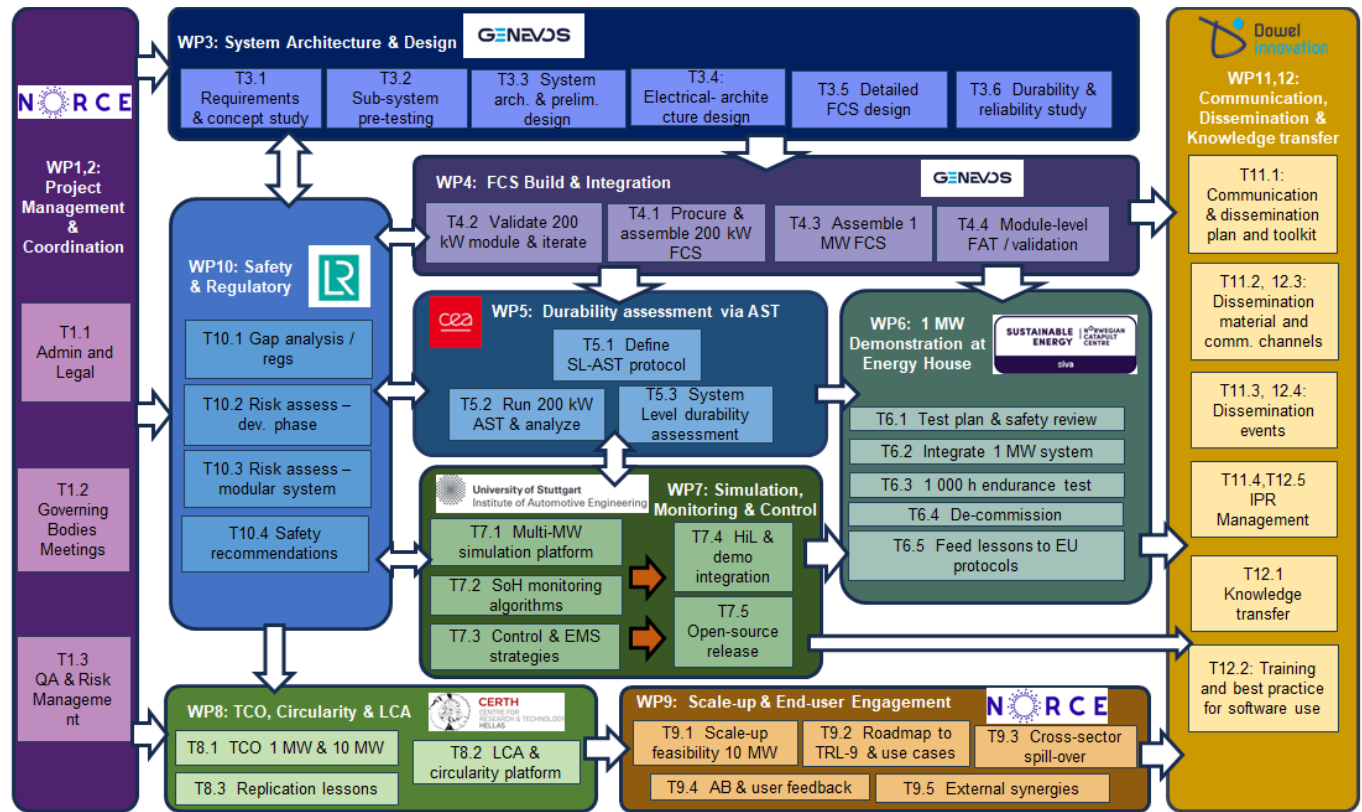
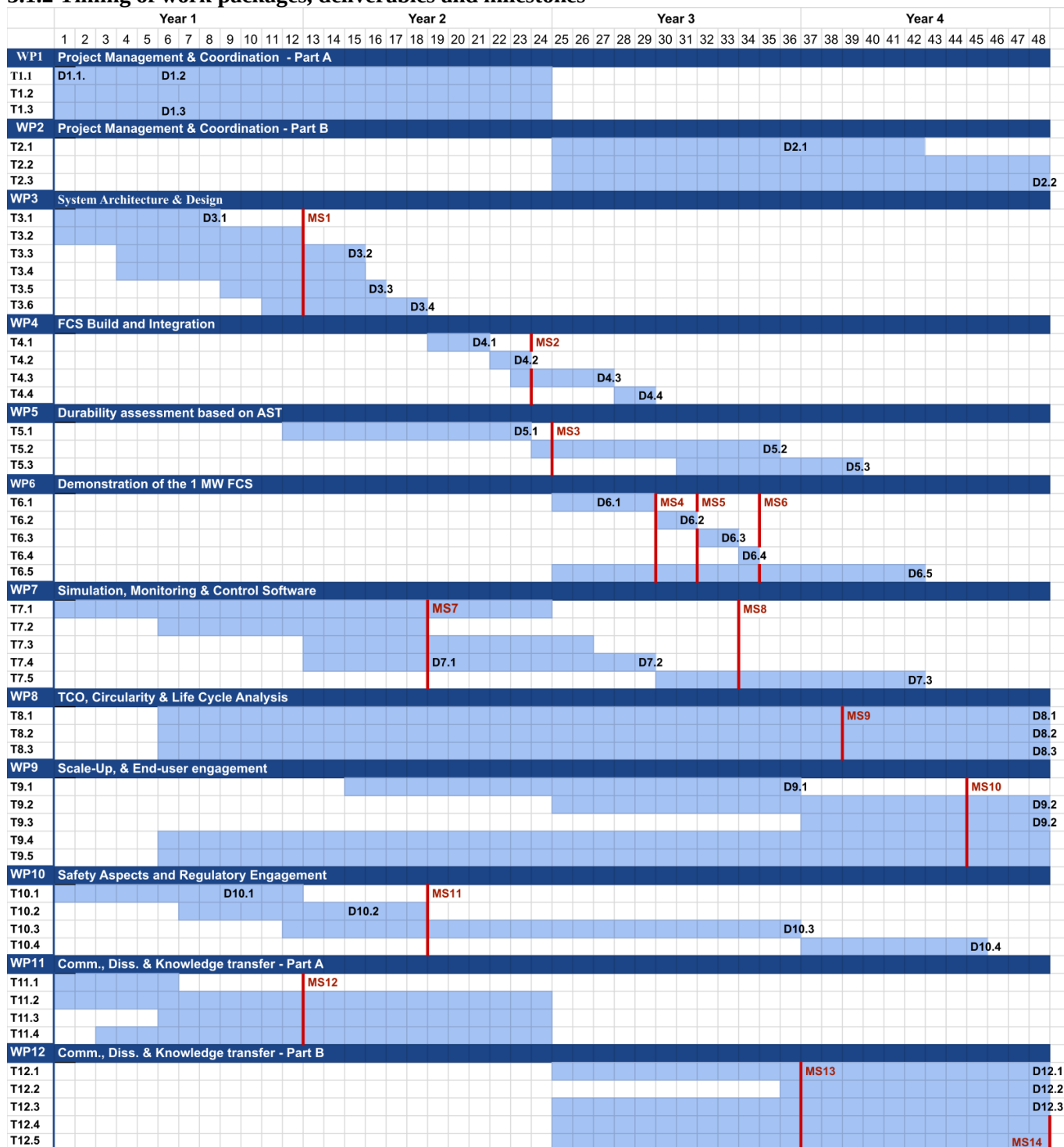


Figure 8: Pert diagram of the work plan.

Table 1 - List of work packages

WP #	WP Title		Lead	PMs	Start	End
1	Project Management & Coordination – Part A	1	NORCE		M1	M24
2	Project Management & Coordination – Part B	1	NORCE		M25	M48
3	System Architecture & Design	5	GENEVOS		M1	M18
4	FCS Build and Integration	5	GENEVOS		M19	M29
5	Durability Assessment based on AST	13	CEA		M12	M39
6	1 MW 1000 Hour Demonstration	12	SEC		M25	M42
7	Simulation, Monitoring & Control Software	6	USTUTT		M1	M42
8	Assessment of MARINER solution	10	CERTH		M6	M48
9	Scale-Up & End-User Engagement	1	NORCE		M6	M48
10	Safety Assessment & Regulatory Engagement	8	LR EMEA		M1	M48
11	Communication, Dissemination & Knowledge Transfer A	11	DOWEL		M1	M24
12	Communication, Dissemination & Knowledge Transfer B	11	DOWEL		M25	M48

3.1.2 Timing of work packages, deliverables and milestones



3.1.3 Work Package Descriptions

WP Number	1	Lead Beneficiary	NORCE	Start month	M1	End Month	M24
WP title	Project Management & Coordination - Part A						
OBJECTIVES							
[1] NORCE coordinates and manages the project at a strategic level, ensuring effective execution of all activities and timely reporting to the EC; [2] establishes and maintains the administrative, legal, and financial framework for the consortium (consortium agreement, IPR issues, budget management); [3] provides clear governance structures and decision-making processes via regular Management Committee meetings; [4] oversees quality assurance and risk management (technical, financial), putting in place a monitoring system for deliverables and project milestones;							

[5] maintains effective communication channels among consortium partners, and manage interactions with the EC.

DESCRIPTION OF WORK

T1.1: Admin and Legal (Lead: NORCE, Participants: All) (M1–M24):

Subtask (ST) 1.1.1 Procedures: The decision-making structure is organised around the project coordinator (PC), WP leaders, and the Technical Committee. NORCE will develop and implement the project's administrative and legal framework, including drafting the consortium agreement, handling IPR arrangements, and ensuring compliance with EU guidelines. A Data Management Plan (DMP) will be defined and kept up to date through the project as a living document, updated every year or when project results require it. The DMP will ensure the fulfilment of EU legislation on privacy issues, IPR and data management.

ST 1.1.2 Administrative assistance: This involves (i) coordinating, managing, and supporting administrative matters; (ii) compilation of deliverables; (iii) periodic reporting and communication with EC and Project Officer(s) (PO); and (iv) progress monitoring and reporting.

T1.2: Governing Bodies Meetings (L: NORCE, P: All) (M1–M24):

subtask (ST) 1.2.1. Project management meetings: (i) Kick-off meeting (M1); (ii) General assembly face-to-face meetings (M16, M32) and online meetings every other month; and (iii) Final meeting (M48). *ST 1.2.2. Technical Committee Meetings:* The TC will meet once a year.

T1.3: QA & Risk Management (L: NORCE, P: All) (M1–M24): Under this task, NORCE will establish a quality assurance plan and a risk management framework. QA activities include deliverable peer-review, milestone verification, and compliance checks with agreed-upon standards. A Risk Register will identify potential issues (technical or financial), define mitigation measures, and track their resolution. NORCE will produce updates every 6 months on the project's risk status, ensuring any deviations are swiftly addressed.

WP Number	2	Lead Beneficiary	NORCE	Start month	M25	End Month	M48
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WP title Project Management & Coordination - Part B

OBJECTIVES

Building on the structures and procedures set up in **WP1**, WP2 continues the same objectives and tasks from M25 to M48.

DESCRIPTION OF WORK

T2.1: Admin and Legal (L: NORCE, P: All) (M25–M36)

T2.2: Governing Bodies Meetings (L: NORCE, P: All) (M25–M48)

T2.3: QA & Risk Management (L: NORCE, P: All) (M25–M48)

WP Number	3	Lead Beneficiary	GENEVOS	Start month	M1	End Month	M18
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WP title System Architecture & Design

OBJECTIVES

Under WP3 GENEVOS and the consortium partners will define, simulate, and detail the complete fuel cell system design for the 200 kW and the 1 MW fuel cell system. GENEVOS will rely on close collaboration with the end users, class society and development partners in order to define and develop a modular fuel cell system design that ensures high flexibility of integration with different vessel types, high scalability, fuel reformer compatibility, as well as setting new standards in operating efficiency and operational lifetime. The design requirements and electrical architecture will be heavily driven through key identified market applications, requiring high power systems up to 10MW, for which end user partners and class will have a valuable involvement. Stack characterisation will be carried out by MINES in order to identify preferential operating points to optimise longevity and efficiency, which will heavily drive the fuel cell system BOP design concept. MINES will also implement modelling and simulation of the system to optimise the design of the multi-stack architecture, BOP, and scalability in relation to technology availability, sizing and readiness. The complete design documentation for FCS units 200 kW and 1 MW will be developed by GENEVOS together with the electrical architecture definition to ensure scalability, flexibility and standardization (StasHH), ensuring class compliance, and importantly confirming the reliability and safety mechanisms for maritime applications.

DESCRIPTION OF WORK

T3.1: Requirements & Concept study (L: GENEVOS, P: LR EMEA, VARD, NORCE) (M1-M8): GENEVOS will determine the design requirements for a scalable fuel cell system tailored to ship applications, considering user constraints, class requirements, and state-of-the-art technologies. It involves a detailed analysis of end-user needs

and ship work profiles, alignment with marine safety regulations for fuel cell systems and hydrogen reformers, and research on existing high-power fuel cell designs and standards, including StasHH references. GENEVOS with VARD and NORCE will identify ship integration constraints, perform trade-off analyses for different installation concepts, and together with LR EMEA, develop a safety strategy in line with class society and marine safety regulations. It culminates in defining and documenting conceptual designs for 1+ MW systems, aligning installation concepts with regulatory requirements, and generating a comprehensive Design Requirements document for the fuel cell system.

T3.2: Subsystem Pretesting of Concepts (L: MINES P: CEA, VARD) (M1-M13): MINES will facilitate model-based system architecture design and integration through early testing and characterization of key components, focusing on the fuel cell stack and air filtering system. The goal is to optimize operating points for maximal efficiency and durability. The process begins with characterizing a representative short stack (< 5 kW) under various conditions, including current, temperature, pressure, stoichiometry, and relative humidities, using polarization curves and step responses. Humidity measurements and water balance assessments will be conducted. Load cycle tests will be conducted to gather degradation data over 400 hours. Additionally, the task involves characterizing the air filtering system under marine conditions to mitigate contamination risks from sea salt and pollutants, supporting the sizing and maintenance strategies for the system.

T3.3: FCS System Architecture & Preliminary Design (L: GENEVOS, P: MINES, VARD, LR EMEA) (M4-M15): GENEVOS develops and optimizes the core fuel cell system architecture, including stack configuration, air filtration, cooling, heat recovery, and electrical architecture, using design requirements and subsystem test results. GENEVOS together with MINES will identify and incorporate recent research and advances in stack architecture to ensure state-of-the-art design, building on the characterization from previous tasks. The process involves high-level design studies, technology reviews, and cost analyses to drive system concept and modularity. It includes numerical modeling to simulate and optimize system operating conditions for efficiency and durability, defining the fuel cell system module architecture, and creating a bill of materials. Mechanical design concepts, interface documentation, and safety strategies are developed, with validation from Class societies. The task also identifies interconnection protocols and physical constraints, with active participation from LR EMEA and other partners to address safety and regulatory aspects, feeding into broader safety and risk assessments. Results of WP3.3 will feed in WP10.2 for safety assessment, risk assessment studies and regulatory aspects.

T3.4: Optimized electrical architecture design for the modular system (L: VARD, P: GENEVOS, LR EMEA) (M4-M15): VARD designs and models the complete power-conversion chain for both the 200kW module and the 1 MW demonstrator using marine-rated commercial-off-the-shelf (COTS) DC/DC converter bricks. We begin with a market and standards benchmark of 100–300 kW bricks, ranking them on efficiency, footprint, cost, lead-time and class certification. We develop the optimization methodology to define the optimal power conversion architecture. A common DC-bus window, fully compatible with StasHH electrical interfaces, is frozen, after which we design a modular connection scheme that supports hot-swap, N-1 redundancy, graceful degradation and integrated protection (fusing, isolation, earthing, EMC). LR EMEA reviews and validates the resulting optimized design and preliminary FMEA to address requirements of marine safety rules. Two shortlisted bricks are then subjected to qualification tests at CEA—efficiency, transient response, fault ride-through, parallel operation and thermal imaging—under moist-salty environmental conditions. Finally, we produce the mechanical layout, cooling concept and interface-control documents for the 200kW module and the containerised 1 MW system.

T3.5: FCS Detailed Design (L: GENEVOS, P: VARD, LR EMEA) (M9-M16): GENEVOS will perform the detailed design and specification of the 200 kW A-Proto and full 1 MW fuel cell systems, utilizing outcomes from previous tasks. The design of the 200 kW A-Proto will be assembled and validated first, allowing improvements and test bench upgrades to be incorporated into the final 1 MW system design. GENEVOS will define a complete Bill of Materials (BoM) for both systems, controlling and validating BoM components, and conducting thermal and fluid simulations on the final design. It includes completing CAD designs for the 200 kW and 1 MW systems, optimizing layouts for maintenance accessibility, and designing test bench installations with electrical interfacing. Assembly drawings and design documentation will be prepared, with design reviews and updates integrated following the validation and improvements from the 200 kW tests.

T3.6: Durability and Reliability Assessment (L: GENEVOS, P: LR EMEA, VARD) (M11-M18): GENEVOS defines reliability targets for the fuel cell system to achieve 40k hours in design and 80k hours in feasibility, aligning with EU 2030 targets at both component and system levels. It involves conducting a Failure Modes and Effects Analysis (FMEA), together with LR EMEA, to identify potential failure points and developing redundancy and fail-safe strategies for single and multi-stack configurations. GENEVOS and LR EMEA incorporate maritime environment stress factors such as vibration, humidity, and salinity to ensure robust performance. It includes establishing safety shutdown protocols and compliance requirements, validating these measures against the

system's architecture, Balance of Plant (BoP), and detailed design. The outcomes are documented, listing all reliability measures and safety features to ensure comprehensive system reliability and safety.

WP Number	4	Lead Beneficiary	GENEVOS	Start month	19	End Month	30
WP title	FCS Build and Integration						
OBJECTIVES							
The goal of WP4 is the [1] physical procurement, [2] assembly and [3] validation of the modular Fuel Cell System (FCS) developed in WP3. It covers all steps from component procurement and assembly of the 200 kW prototype to the full 1 MW FCS integration, while ensuring compliance with pre-standards safety protocols for maritime deployment.							
DESCRIPTION OF WORK							
<u>T4.1: Component Procurement & Assembly of 200 kW FCS (L: GENEVOS, P: VARD) (M19-M21):</u> GENEVOS and VARD initiates the procurement of necessary components for the assembly of the 200 kW A-Proto as soon as its detailed design is complete. It involves finalizing supplier procurement specifications based on the validated Bill of Materials (BoM) from WP3, followed by the procurement of the entire BoM according to build schedules and lead times. The task includes supplier follow-ups to ensure quality, cost, and timely delivery, as well as component and subassembly validation testing for systems such as cooling, air conditioning, safety, and control. These steps aim to de-risk full-scale testing and reduce debugging time, culminating in the full assembly of the 200 kW FCS A-Proto module.							
<u>T4.2: 200 kW FCS Validation Testing & Development (L: GENEVOS, P: VARD, LR EMEA, CEA) (M22-M23):</u> GENEVOS and CEA validate the functional, performance, and safety-related aspects with input from LR EMEA, of the Fuel Cell System (FCS) design at the module level before the AST. It involves defining test procedures for functional, safety, performance, and environmental assessments, and conducting lab-based functional validation testing of the 200kW system. GENEVOS will analyze and optimize system operating conditions to meet design targets for efficiency and durability, with Factory Acceptance Testing (FAT) conducted together with VARD and LR EMEA according to Class requirements. Design improvements and hardware/software upgrades are identified and incorporated into the 1 MW system design prior to assembly, with iterative updates made where possible to ensure a reliable and high-efficient system.							
<u>T4.3: Assembly of 1 MW FCS (L: GENEVOS, P: VARD) (M23-M27):</u> Design freeze, final procurement and assembly of full high power system modules followed by integration in a container with power electronics and external interfaces for test bench. GENEVOS finalizes the design of the 1 MW system based on insights from the 200 kW A-Proto testing, followed by the procurement of any remaining or replacement components. GENEVOS will assemble the 1 MW system modules, encompassing mechanics, fluid, control, and communication systems, and documente the assembly process. VARD and GENEVOS will then integrate FCS modules and power electronics into a container enclosure, following Factory Acceptance Testing (FAT) of the modules. Finally, initial system-level factory checks are conducted (mechanical, fluid, and electrical).							
<u>T4.4: Modular FCS Validation & FAT (L: CEA, P: GENEVOS, LR EMEA) (M28-M29):</u> Prior to assembling the full containerised 1 MW system and sending to Norway, modules will be validated individually at CEA Nantes 500 kW lab, through class-approved FAT protocol, but also with the objective to parallel two modules to validate the high-power electrical architecture. The process includes defining FAT protocols based on class recommendations and end-user profiles, performing subsystem and subfunction testing for each module independently, and conducting integrated module-level FAT for communication, control, and safety. Key performance characteristics such as efficiency, maximum power, and ramp response are validated at the module level. Additionally, a multi-module test is conducted with at least two FCS modules to validate parallel connections, insulation resistance, independent controllability, redundancy, and system efficiency.							

WP Number	5	Lead Beneficiary	CEA	Start month	M12	End Month	M39
WP title	Durability assessment based on AST						
OBJECTIVES							
The aim of WP5 is to define and implement a testing protocol and methodology for assessing reliability and durability of high-power FCS dedicated to maritime applications. This methodology will be based on system-level Accelerating Stress Testing (AST) coupled with data-driven models for the extrapolation of testing results according to real load profiles and scaling (from 1 to 10 MW). A particular attention will be paid to the reproduction of maritime environmental conditions for the definition of testing protocols.							

DESCRIPTION OF WORK	
Task 5.1: Extended-AST protocol definition (L: MINES, P: CEA, LR EMEA) (M12-M23): CEA and MINES defines an AST protocol adapted to fuel cell systems in maritime applications. Starting from the background in AST definition and implementation for fuel cells and the requirements for maritime use case, a model-based approach will be able to quantify main influencing conditions on fuel cell durability (environmental conditions, cycling load profile, stop and start, etc.). Physics-based aging models calibrated with preliminary data from T3.2 will be used initially to quantify those aging conditions and define a protocol enabling aging acceleration by a factor between 20 and 40. We will perform a total of 400 h of preliminary short-stack (< 5 kW) tests with one or more candidate AST protocols. Those tests will validate that the chosen AST protocol is representative of the degradations induced by the real operation of the FCS. MINES and CEA will collaborate in defining this system level AST protocol by using physical models and the results of short-stack tests performed by MINES. LR EMEA will contribute in reviewing the protocol considering current requirements for fuel cell certification.	
Task 5.2: Module level AST implementation and analysis (L: CEA, P: MINES, GENEVOS) (M24-M35): CEA will implement the AST protocol defined in the previous task at module scale (200 kW) with the support of GENEVOS. We will use a climatic chamber with moist conditions, salty air inlet and consider different air inlet temperatures (to simulate different installation areas on board of vessels). Different dynamic load profiles will be applied in a simulation approach emulating the overall ship power demand. The AST will be applied on the FCS module, including its BoP components. Testing results will be used to develop and set-up data-driven surrogate models of fuel cell aging in collaboration between CEA and MINES. Starting from physical-based models used in the previous task, the aim will be to evaluate an hybrid data-based approach to improve accuracy and robustness enabling results extrapolation.	
Task 5.3: System level AST results extrapolation and durability assessment (L: CEA, P: MINES, LR EMEA) (M31-M39): With data-based aging models tuned with AST results, CEA will extrapolate real testing results for assessing the durability and lifetime of the system (up to 40 000h) considering different use cases and corresponding load profiles. This task will finally provide a synthesis of methodology means and results (protocols, testing facilities, data and models) for maritime high-power FCS and will assess how this methodology can improve the design and integration on board of high-power FC systems, particularly with regard to robustness and certification constraints.	

WP Number	6	Lead Beneficiary	SEC	Start month	M25	End Month	M42
WP title	Demonstration and Validation of the 1 MW Fuel Cell System at Energy House						
OBJECTIVES							
To demonstrate and validate the performance, efficiency, reliability, and safety of a 1 MW hydrogen fuel cell system (FCS) under controlled, simulated maritime conditions at the Energy House full-scale test infrastructure. This includes the integration of containerised FCS modules into the facility, the implementation of relevant test protocols, and the generation of operational data to support future certification, scaling and maritime deployment.							
DESCRIPTION OF WORK							
Task 6.1: Development of test plan for Energy House demonstration.: (L:VARD, P: GENEVOS, SEC, LR EMEA) (M25-M29): SEC, VARD, and GENEVOS design a comprehensive test plan that meets both project objectives and regulatory expectations for 1 MW maritime fuel cell systems. Test scenarios will include simulated maritime conditions such as salt-laden air, varying humidity, and temperature differentials representing different vessel installation environments. The test plan will detail load profiles emulating ship operation, failure mode evaluations, and emergency handling protocols, and will be co-developed with system developers to ensure coverage of all critical design aspects. The plan will also identify instrumentation requirements and data collection strategies. LR will review the proposed test schedules and provide recommendations based on marine fuel cell system testing requirements, reliability and dependability considerations as well as the need for blackout and dead ship recovery.							
Task 6.2: Integration of fuel cell system into Energy House infrastructure (L: SEC, P: Vard, GENEVOS) (M30-M31): In this task, the 1 MW fuel cell system will be physically delivered to the Energy House in containerised format and integrated into the facility's test infrastructure by SEC and VARD. SEC will coordinate all work related to physical installation, connection to the grid and auxiliary systems (cooling, safety systems, control interfaces), and verification of operational compatibility. If required, modifications to the facility’s layout or supporting infrastructure will be performed to enable safe and effective testing. Following installation and system verification, a full commissioning process will be carried out. This includes functional testing under load conditions.							

validation of control systems, and confirmation that the fuel cell system performs in accordance with the Energy House Test Plan. All safety systems will be tested prior to commencement of operation, and documentation covering system layout, interfaces, and emergency procedures will be produced.

Task 6.3: Execution of system validation testing at Energy House: (L: GENEVOS, P: Vard, SEC, NORCE) (M32-M33): Following integration, SEC will test the fuel cell system at full scale under simulated maritime operating conditions for a minimum of 1,000 hours. The testing will involve dynamic loading based on emulated vessel propulsion and auxiliary profiles. Particular attention will be given to evaluating power response, system stability, control algorithms, degradation patterns, and operational resilience. Data will be collected continuously, including thermal, electrical, and diagnostic outputs. Results will be analysed against the initial test plan and KPIs from the Clean Hydrogen JU SRIA including evaluation of net efficiency and degradation, validating the AST model at full scale. Any critical findings will trigger feedback loops for system optimisation.

Task 6.4: Decommissioning of 1 MW fuel cell system from Energy House: (L: SEC, P: GENEVOS, Vard) (M34): This task covers the planned decommissioning and removal of the 1 MW fuel cell system from the Energy House facility at the end of the testing period. SEC will lead the coordination of all decommissioning activities to ensure a safe and structured process in accordance with applicable safety standards and site procedures. Key activities include shutdown and safe isolation of the fuel cell system, including disconnection from electrical, cooling, control, and safety systems; inspection and verification of system condition prior to disassembly, to document operational status and identify any issues arising from test operation; physical disassembly and removal of the containerised fuel cell unit and associated auxiliary components; restoration of the Energy House test area to its pre-installation state, or to a state ready for future test campaigns, depending on project needs; documentation covering decommissioning procedure and timeline, disconnection and removal processes, safety verifications and risk mitigations; handover of equipment to GENEVOS or designated partner. All work will be carried out with a focus on health, safety, and environmental (HSE) considerations, and in close coordination with facility management at Energy House.

Task 6.5: Harmonized testing protocols for performance and durability assessment (L: CEA, P: SEC, Mines, Genevos) (M24-M42): Partners of the Task will collaborate with JRC on the topic of test protocol definition (AST in WP5 and 1 MW testing in WP6) to contribute to EU-wide testing protocol harmonisation. The way this collaboration mechanism will be implemented will be discussed with JRC at the beginning of the Task to fit their preference and could be e.g. via online workshops. Interaction will take place at key steps during the protocol definition and later in the testing and post-test phase.

WP Number	7	Lead Beneficiary	USTUTT	Start month	M1	End Month	M42
WP title	Simulation, Monitoring & Control Software						
OBJECTIVES							
[1] To Develop and integrate advanced PEMFC models (digital twins) that cover performance and degradation at cell, stack and system level; [2] develop physics based and data driven monitoring algorithms with diagnostics for state-of-health (SoH) assessment; [3] deliver health-conscious control and multi-scale energy-management strategies that can be executed in SiL/HiL environments and on the MARINER hardware demonstrators, and [4] to release the software as open-source to foster replication and spill-over.							
DESCRIPTION OF WORK							
T7.1 Multi MW simulation software (L: MINES, P: VARD, USTUTT) (M1-M24): MINES will adapt the open-source simulation platform VirtualFCS for multi-MW fuel cell systems and enhance it with both physics-based and data-driven state-of-the-art fuel cell models, including degradation modelling. Models will also be packed into functional mock-up interfaces (FMI), allowing for flexible use in various existing simulation tools. An example case for simulating a multi-stack FCS for maritime applications will be added to VirtualFCS.							
T7.2: Monitoring Algorithms (L: MINES, P: USTUTT, LR EMEA, VARD) (M6-M18): MINES will implement a multi-level SoH monitoring framework using both physical measurements and models and data-driven analytics. To anticipate potential degradation while keeping improving performance, a particular attention will be given to the monitoring of the FC membranes humidity level, which are not directly measurable and should be recovered by alternative means. This includes the design of 1) model-based humidity observers and 2) Machine Learning methods based on the outputs of T.7.1. These strategies will enable early detection of aging or performance degradation as well as anomalies at the module and stack level, to maintain safe operation. They can also be used to develop SoH-aware control algorithms to extend system lifetime.							
T7.3: Control Algorithms (L: MINES, P: CEA, USTUTT, LR EMEA, VARD) (M13-M26): MINES design advanced real-time control algorithms and Energy Management Systems, maximizing System efficiency and							

durability through enhanced State of Health (SoH) awareness from hybrid degradation models (T7.1) extrapolating the results for 1 MW for different use cases (WP 5). These control algorithms will exploit SoH monitoring algorithms developed in T.7.3 and simulation tools developed in T.7.1. The resulting control strategies will have to be compliant with various systems architecture, depending on the outcome of WP3, and thus enable flexibility and modularity. LR EMEA will review the control and monitoring software outcomes regarding safety related impacts on system and ship.

ST 7.3.1: *Dynamic pressure and flow control of air and hydrogen:* This subtask will develop a comprehensive control framework for innovative pressure and flow control of air and hydrogen, and integrated auxiliary systems (e.g. humidifiers, pumps, ejectors) based on both system state and predicted degradation trends, to ensure optimal and smooth operation for each stack. Cooling strategies will be adapted accordingly, to control the humidity level. Leveraging on the simulation tools developed in T.7.A, the resulting control strategies will mitigate stress and fatigue of the critical air and hydrogen loop components, maximizing the system lifetime and durability.

ST 7.3.2: *Recirculation loop control and purge strategy:* This subtask will focus on recirculation control to manage hydrogen utilization and improve efficiency, while mitigating water accumulation and membrane drying. A particular attention will be given on purge strategy optimization, particularly for nitrogen and water management, using real-time sensor feedback and SoH estimates to trigger purging only when necessary—thus minimizing hydrogen loss and improving autonomy.

ST 7.3.3: *Multi-Scale Energy Management Systems for Independent Module/Stack Control:* This subtask will introduce a multi-scale Fuel-Cell Management System, designed to adapt dynamically to the Modules/Stacks monitoring. At the top level, this Management System (MS) will distribute the power demand between the battery management system and the FC, to jointly operate them in an optimal manner. At the FC level, this MS will independently control each module's power output, thermal management, and safety status and optimize load sharing to select suitable current density for each stack or module, based on real-time SoH data and predicted degradation. In addition, it will manage failure cases by dynamically reallocating power, ensuring partial operation even during faults. This modular EMS will be tested on a Hardware-in-the-Loop (HiL) testing platform at CEA, specifically designed for prototyping modular fuel cell architectures.

T7.4: Integration of software into test system (L: USTUTT, P: MINES, GENEVOS, CEA) (M13-M29): USTUTT port the T7.1-T7.3 toolchain to the CEA HiL bench and to the 1 MW demonstrator control stack. Validate transient response, fault-ride-through and SoH-aware load sharing under design and off-design scenarios. Results are fed back to WP4/6 for firmware freeze.

T7.5: Open-source publishing (L: USTUTT, P: MINES) (M30-M42): USTUTT Package the full code base (models, diagnostics, control, EMS) under a permissive licence, generate API documentation, tutorials and a Docker image, and deposit it on GitHub + Zenodo with a DOI. Host a public webinar and contribute a training module to WP12.

WP Number	8	Lead Beneficiary	CERTH	Start month	M6	End Month	M48
WP title	Assessment of MARINER solution						
OBJECTIVES							
[1] To assess the total cost of ownership (TCO) of the MARINER solution for both the demonstration and scale-up phases. [2] To conduct an environmental assessment of the MARINER solution using a life cycle thinking approach. A life cycle inventory will be developed based on data provided by MARINER stakeholders. [3] To develop a digital platform enabling environmental and economic assessments across a range of use cases, supporting replication of the MARINER solution. [4] To compile a set of upscaling guidelines, designed for end users and ship owners, to facilitate broader implementation of the MARINER solution.							
DESCRIPTION OF WORK							
T8.1: Analysis of Total Cost of ownership for demo case (1 MW) and up-scale (10MW) (L: CERTH, P: VARD) (M6-M48): CERTH will conduct a comprehensive analysis of Total Cost of ownership (TCO) based on Life Cycle Thinking approach (LCC), in order to evaluate the cost-effectiveness of all processes associated with the MARINER demonstration case (1 MW) and scale-up scenario (10 MW). Capital expenditure (CAPEX), including infrastructure and equipment costs, will be estimated based on input from technology providers. Operating expenditure (OPEX) will be assessed in alignment with the use cases defined in WP9. The TCO analysis will identify key economic indicators and cost drivers, supporting the economic feasibility assessment of MARINER. Additionally, a comparative cost analysis with alternative decarbonization technologies in the maritime sector will be carried out to assess MARINER's market competitiveness and potential for sectoral uptake.							
T8.2: Life Cycle Assessment Approach integrated with Circularity Design under a digitalized platform (L:							

CERTH, P: VARD, LR EMEA) (M6-M48): CERTH develop a circular economy-driven methodological framework for the sustainable design, manufacturing, and operation of the MARINER Fuel Cell System (FCS). The approach will emphasize circularity principles, with particular attention to end-of-life recyclability and material recovery. The life cycle assessment (LCA) will combine attributional and consequential approaches and align with ISO 14044 standards, as well as the European Union's sustainability and circularity goals for hydrogen systems. System boundaries will encompass the infrastructure, operational processes, and end-of-life phase. Environmental performance will be evaluated for both demonstration and scale-up cases, in line with WP9. Comparative LCA will be conducted against alternative powertrain technologies—such as battery systems, biofuels, and Solid Oxide Fuel Cells (SOFCs). A dedicated digital tool will be developed to enable dynamic assessments of environmental and economic performance across various configurations and demand profiles, as defined by relevant use cases.

T8.3: Lessons learnt for replication (L: CERTH, P: NJORD, KIARA) (M24-M48): CERTH and NJORD will develop a comprehensive set of guidelines based on TCO data and LCA to support the upscaling and replication of the MARINER solution. Insights from the circularity design, system testing, and demonstration activities will inform the development of practical rules and best practices to support scale-up. The guidelines will address vessel type (e.g., size, configuration) and route variability to ensure wide applicability. Replication opportunities will be mapped, outlining how MARINER can be deployed across different maritime contexts. Task 9.1 will provide feedback to this task.

WP Number	9	Lead Beneficiary	NORCE	Start month	6	End Month	48
WP title	Scale-Up Feasibility & End-user engagement						
OBJECTIVES							
The objective of this work package is to define and validate the best approach to upscale and implement the proposed FCS to 10 MW scale (and beyond) based on real end-user scenarios, develop a roadmap to TRL9 together with stakeholders and end-users and assess potential for the spillover of the technology to other use cases. The work package will focus on: [1] Evaluating the feasibility of scaling from 1 MW to 10 MW using simulation and cost analysis data; [2] Developing a roadmap from TRL6 to TRL9 with concrete end-user scenarios in collaboration with a ship operator; [3] Assessing modifications for cross-sector applications (spillover). [4] Fostering stakeholder engagement through advisory board meetings, workshops, and best-practice guidelines. [5] Strengthening external partnerships with key maritime fuel cell projects and European initiatives.							
DESCRIPTION OF WORK							
T9.1: Scale-Up Feasibility (1 MW → 10 MW) (L: NORCE, P: CEA, VARD, SCORPIO) (M15-M36): NORCE outline expansions; synergy with FCS numerical models & TCO data for multi-MW design. The outputs of T3.5, 3.6 and 5.3 will serve as the foundation for assessment of the scale up to 10 MW, when the assembly and sizing of components of the FCS will be analyzed. We aim to predict how modifications in the architecture might affect overall performance. We will assess whether any components or systems (e.g., BoP elements, power converters, or cooling systems) serve as enablers for scale-up or could present bottlenecks in terms of performance, durability, or efficiency. Incorporating TCO and LCA guidelines from T8.3, the reliability and lifetime projections, market and operational data into the analysis will link technical performance with economic viability. We will establish a clear view into the main cost drivers of the FCS (capital, operational, maintenance as well as regulatory and compliance costs) in close collaboration with the end-users and evaluate cost - performance design trade-offs to ensure the scaleup strategy is economically sound for the most cost-competitive and reliable future product.							
T9.2: Roadmap to TRL9 & Use Case (L: VARD, P: NORCE, DOWEL, SCORPIO, KIARA, NJORD) (M25-M48): Two end-user scenarios (a passenger ferry [KIARA] and a tanker [Scorpio Group]) will be used, together with green retrofit experts NJORD, to develop a detailed roadmap for advancing the TRL of the FCS from 6 to 9. This roadmap will detail technical milestones and validation steps, integrating pilot demonstrations and simulation-based assessments. It will include proactive risk mitigation—particularly in certification and type approval—to avoid delays and secure a market-ready, compliant technology. Feedback from stakeholders and end users (T9.4 and 9.5) will be actively sought and used to improve the value of this Roadmap.							
T9.3: Investigate Spillover (L: NORCE, P: GENEVOS) (M37-M48): NORCE will investigate spillover by assessing how the modular, scalable design of the FCS may be adapted for applications such as trains, trucks, buses, heavy-duty vehicles, stationary power, rail systems, off-grid or microgrid solutions, port and harbor operations, and industrial backup systems. We will compare operational requirements and performance challenges across these sectors and evaluate the economic viability of implementing FCS technology on a case-by-case basis. Additionally, we will assess the standardization and certification requirements relevant to these areas, and how differences in							

regulatory frameworks may affect or inform the potential spillover strategy.

T9.4: Advisory Board & User Engagement: Feedback and Best Practices (L: NORCE, P: DOWEL, VARD, LR EMEA, SCORPIO, KIARA) (M6-M48): NORCE will gather user input at regulatory & non-technical level through yearly meetings with the Stakeholder advisory board and also a Workshop with both the end-users and stakeholders. The feedback and recommendations will be compiled into a best-practice guide ensuring alignment with the market needs and regulations.

T9.5: External Partnerships & Synergies (L: CERTH, P: NORCE, DOWEL) (M6-M48): We will maintain links and ensure relevant communication with H2MARINE and other maritime FC projects (ZEW, StaSHH, HyShip, etc.) through direct contact with project coordinators while also actively monitoring recent publications. Relevant communication will be carried out for synchronisation of our activities. For this reason, a workshop will be organized in order to invite members from networking projects to share their knowledge with the MARINER consortium. In addition CERTH in contact with KIARA will organize workshops inviting industrial stakeholders and policy makers. This approach will enhance the value of our work, synchronise our activities, promote synergy, and likely feed into T9.1 and 9.2.

WP Number	10	Lead Beneficiary	LR EMEA	Start month	1	End Month	48
WP title	Regulatory Aspects and Safety Assessment						
OBJECTIVES							
This work package focuses on ensuring the safety of the fuel cell power system installation on board ships through assessment of the safety requirements for their development and modular arrangement. The main objectives of this work package are [1] Review Regulatory aspects of FC systems and Gap analysis; [2] Safety assessment of Fuel Power Systems and their modular arrangement on board ships; [3] Safety analysis and Recommendations.							
DESCRIPTION OF WORK							
T10.1: Review of Regulatory aspects of FC systems and Gap analysis (L:LR EMEA, P: VARD, GENEVOS) (M1-M12): Review of applicable marine safety standards and class rules for fuel cell power installations. Gap analysis of the proposed design and arrangement, and class rules will be performed to determine if an alternative design approach needs to be selected. Testing requirements for safety validation of fuel cell power systems will be elaborated.							
T10.2: Risk assessment of fuel cell power system development (L: LR EMEA, P: VARD, NORCE, GENEVOS) (M7-M18): The system design proposed as outcome of WP3 and WP4 undergoes evaluation and risk assessments using the LR EMEA’s Risk Based Design process and procedures. For a successful outcome of the risk assessment studies an active participation of the project partners GENEVOS and VARD is essential. The safety assessment includes a thorough review of relevant rules, regulations, guidelines, and standards pertaining to marine fuel cell power systems and their installation on ships. Safety requirements are prioritized early in the project to integrate findings into the system design. Regular reviews are conducted throughout the project to ensure compliance with any new regulations that may arise. The results will be summarized and contributed to the compilation of a safety analysis report for the fuel cell power system design.							
T10.3: Risk assessment of modular arrangement of fuel cell power system (L: LR EMEA, P: VARD, GENEVOS) (M12-M36): This task focuses on the modular arrangement and integration of 1-MW fuel cell power systems on board ships. Risk assessments workshops will be held to assess the risks, hazards and considered safeguards of the installation. The risk assessment studies will be based on the documentation and outcomes of WP3 and WP4. For a successful outcome of the risk assessment studies an active participation of the project partners GENEVOS and VARD is essential. Requirements on system redundancies, including auxiliary systems of the fuel cell power installation, will be investigated. The results will be summarized and contributed to the compilation of a safety analysis report for the fuel cell power system integration on board.							
T10.4: Safety analysis and Recommendations (L: LR EMEA, P: VARD, NORCE, GENEVOS, NJORD) (M37-M45): This task will summarize the learnings from T10.2 and T10.3 and elaborate the requirements for further risk studies as well as rule and standard developments to close regulatory gaps.							

WP Number	11	Lead Beneficiary	DOWEL	Start month	1	End Month	24
WP title	Communication, Dissemination & Knowledge transfer - Part A						
OBJECTIVES							
[1] Maximise the impact of the project activities and outcomes by engaging all relevant stakeholders and fostering the exchange of knowledge and experience; [2] Prepare the exploitation of project results, including intellectual							

property management; [3] Promote the sustainable and widespread use of the MARINER-developed tools within the maritime industry; [4] Contribute to upskilling in industry and enhance SMEs involvement in the hydrogen economy.

DESCRIPTION OF WORK

T11.1: Communication & dissemination plan and toolkit (L: DOWEL, P: all) (M1-M6): This task will kick-start the project's communication and dissemination activities of the project through the development of the *Communication, Dissemination and Exploitation (CDE) Plan*, due at M6. The plan will outline the overall strategy, including target audiences, objectives, key messages, communication channels, and the associated planning and validation procedures. Specific attention will be paid on promoting cross-sector collaboration, reaching out to SMEs, and contributing to the upskilling of industry stakeholders. The CDE&N plan will provide clear guidelines and a concrete action plan to empower all partners to actively engage in communication, dissemination, and knowledge transfer activities. In parallel, a communication toolkit will be delivered, featuring the project's visual identity (M4), website (M6), LinkedIn account, and supporting materials such as templates, a project presentation, and a leaflet (M6).

T11.2: Dissemination material and communication channels (L: DOWEL, P: all) (M1-M24): DOWEL will develop and manage various communication and dissemination outputs to support the visibility and outreach of the project. It includes the design of dissemination materials—such as brochures, posters, or other content—tailored to the consortium's needs and aligned with specific project activities or target audiences, particularly in support of WP9's stakeholder engagement efforts. It also encompasses the coordination and tracking of peer-reviewed publications, ensuring timely online access, as well as the regular update of the project website with validated public deliverables (excluding the publication of Open-Source models, which is addressed under T7.4). Additionally, this task includes the management of the project's online presence, primarily through the LinkedIn account, to foster engagement with the wider community.

T11.3: Dissemination events (L: DOWEL, P: all) (M6-M24): The project will organise dedicated events to present its approach and progress, both online and in person. Those will include a series of webinars, and the organisation of or contribution to dedicated sessions in specific events such as the European Hydrogen Week, or the European Hydrogen Forum. In coordination with Task 9.5, DOWEL will support knowledge exchange and liaison efforts with EU bodies and other maritime consortia, ensuring alignment and complementarity with broader sectoral developments. The task will also coordinate and monitor the contribution of partners to all relevant events, from European to local level, ensuring strategic project representation.

T11.4: IPR Management (L: DOWEL, P: all) (M3-M24): IPR management activities will focus on the identification and protection of exploitable results to facilitate their uptake after the project ends. An IPR strategy will be defined early in the project in collaboration with the consortium to: 1) prevent any conflicts of ownership, and 2) define the open-source strategy, including the selection of appropriate licenses for models and dissemination materials. T11.4 will build upon the draft exploitation plan and ensure that results are integrated into the CDE plan (by M6). Throughout the project, DOWEL will engage partners through at least two exploitation and IPR workshops, as well as bilateral meetings, to: 1) promote best practices in result identification and protection, in line with the Grant and Consortium Agreements, and 2) formalize what results are exploitable, by whom, for whom, and how. The strategy will ensure clarity and consensus on licensing, ownership, and the future use of results.

WP Number	12	Lead Beneficiary	DOWEL	Start month	25	End Month	48
WP title	Communication, Dissemination & Knowledge transfer - Part B						
OBJECTIVES							
Mirror of WP 11.							
DESCRIPTION OF WORK							
T12.1 Knowledge transfer towards industrials and standardisation bodies (L: GENEVOS, P: NORCE, DOWEL) (M25-M48): This task will focus on facilitating knowledge transfer towards industry stakeholders and relevant European Standardisation Organisations to support future scale-up and market uptake of MARINEFUEL solutions. Although project outputs may not be ready for direct contribution to existing standards, key achievements will be shared through presentations, targeted reports, and participation in selected events and workshops. The task will identify standardisation technical committees in CEN and ISO and will maintain open channels to raise awareness and prepare the ground for potential post-project standardisation initiatives. This task will support the broader engagement activities conducted under WP9 by ensuring that key outcomes from the project are communicated effectively to industrial actors, including port operators, shipbuilders, and maritime fuel cell							

suppliers. It will also ensure that knowledge is disseminated in a way that aligns with the work in classification and regulatory frameworks.

T12.2: Training and best practice for software use (L: MINES, P: CEA) (M36-M48): Mines will develop and deliver a two-day training on modelling fuel cell systems with both theory (1 day) and practical work on Virtual FCS (1 day). This training will address issues such as fuel cell degradation modelling and fuel cell control, and will be delivered in at least one session to OEMs, integrators, ship managers and other end users. The session will be organised in person and will provide in-depth guidance on optimal software use, ensuring that users can integrate these tools into their operations for maximum efficiency.

T12.3: Dissemination material and communication channels (L: DOWEL, P: all) (M25-M48): Mirror task of WP11, with a focus on final dissemination material: project result's brochure, packaging of best practices guide; online release of all public deliverables and access to publications.

T12.4: Dissemination events (L: DOWEL, P: all) (M25-M48): Mirror task of WP11 with a focus on the organisation of the project's final event, possibly held back-to-back with an existing event or jointly with some other project(s) if it is deemed to improve its outreach and impact.

T12.5: IPR Management (L: DOWLE, P: all) (M25-M48): Mirror task of WP11 with a focus on the elaboration of the final results' ownership list, and the design of joint ownership agreements if required.

3.1.4 List of Deliverables

Deliverable number, name and short description		W P	Lead	Type ²⁸	Diss. level	Date
D1.1	Project Manual: Guidance for efficient project implementation	1	NORCE	R	SEN	M1
D1.2	Data Management Plan (1st iteration): Data management procedures to comply with GDPR and other applicable regulations	1	TBD	DMP	SEN	M6
D1.3	Quality and risk Management Plan (1st iteration): Plan for quality control to check that contributions are consistent and meet the requirements for deliverables and reports.	1	NORCE	R	SEN	M6
D2.1	Data Management Plan (2nd iteration): the DMP is updated based on conclusions from the testing protocol and demonstration.	2	NORCE	DMP	SEN	M36
D2.2	Quality and risk Management Plan (2nd iteration)	2	NORCE	R	SEN	M48
D3.1	System Requirements & Architecture Specification: Integration concept study & design requirement for 1 MW	3	Genevos	R	SEN	M8
D3.2	System Requirements & Architecture Specification: Preliminary Design report summarising the design evolution, integrating key findings from stack and component characterisation	3	Genevos	R	SEN	M15
D3.3	System Requirements & Architecture Specification: Detailed design package (modular brick 200k and containerized 1 MW) and integration design for 1 MW FCS test bench	3	Genevos	R	CON	M16
D3.4	System Requirements & Architecture Specification: Reliability and Safety Measure report	3	Genevos	R	SEN	M18
D4.1	FCS Build and Integration: Component Procurement Report & Integration Plan	4	Genevos	R	CON	M21
D4.2	FCS Build and Integration, 200 kW prototype: Assembled 200 kW FCS Prototype with Assembly Documentation and Validation Test Report with Key development points and design upgrades.	4	Genevos	R	SEN	M23
D4.3	FCS Build and Integration, 1 MW assembled: 1 MW FCS Assembly Report detailing fabrication process	4	Genevos	DEM	SEN	M27
D4.4	FCS Build and Integration, 1 MW validated: FCS Validation Test Report at modular and multi-module level with FAT documentation	4	Genevos	R	SEN	M30
D5.1	Accelerated Stress Test Protocol & Methodology: Final procedure for a 40,000-hour (equivalent) AST on at least 200 kW, including modeling to project long-term durability.	5	MINES	R	PU	M23
D5.2	FCS module AST test dataset and updated hybrid FC degradation	5	CEA	R	SEN	M35

²⁸R: Document, report (excluding the periodic and final reports); DEM: Demonstrator, pilot, prototype, plan designs; DEC: Websites, patents filing, press & media actions, videos, etc; DATA: Data sets, microdata, etc; DMP: Data management plan; ETHICS: Deliverables related to ethics issues. SECURITY: Deliverables related to security issues OTHER (OTH) : Software, technical diagram, algorithms, models, etc.

	model: Dataset and updated hybrid model based on 40,000-hour AST of the 200 kW module, supporting lifetime prediction and degradation analysis.					
D5.3	Durability assessment of FCS for different maritime use cases: Report with analysis of expected FCS durability across various maritime profiles using the updated degradation model from D5.2.	5	CEA	R	SEN	M39
D6.1	Energy House Test Plan: Detailed plan outlining test protocols, procedures, and safety measures for validating the 1 MW fuel cell system under simulated maritime conditions.	6	VARD	R	PU	M27
D6.2	Integration and commissioning report of 1 MW FCS at Energy House: Documentation of the installation, connection, and readiness verification of the fuel cell system at the Energy House facility.	6	SEC	R	SEN	M31
D6.3	Validation Report with Performance and Test Data: Comprehensive report of test results, including system performance, reliability, and outcomes from accelerated stress testing.	6	MINES	R	SEN	M33
D6.4	Decommissioning Report of 1 MW FCS at Energy House	6	SEC	R	SEN	M34
D6.5	Lessons Learnt Report: Key lessons learnt and recommendations for harmonized testing protocols	6	CEA	R	PU	M42
D7.1	1 MW simulation model: VirtualFCS extended into a simulation model covering 1 MW FCS.	7	USTUTT	OTH	SEN	M19
D7.2	HiL integration & validation report: Verification results of the software stack on HiL benches and the 1 MW demonstrator.	7	USTUTT	OTH	SEN	M29
D7.3	Open-Source Simulation & Control Toolkit: Public release of multi-MW simulation environment and control software, with documentation for broader maritime adoption.	7	USTUTT	OTH	PU	M42
D8.1	TCO assessment of MARINER solution based on life cycle costing: Detailed report describing methodology and results on TCO assessment	8	CERTH	R	CO	M48
D8.2	Environmental assessment of MARINER solutions based on life cycle thinking: Detailed report describing methodology and results on life-cycle analysis.	8	CERTH	R	CO	M48
D8.3	Guidelines for replication rules based on environmental, economic and technical criteria	8	CERTH	R	PU	M48
D9.1	Technical and economic assessment of module replication to 10 MW system: Detailed report containing the results of the technical and economic analysis of the scale-up of FCS to 10 MW.	9	NORCE	R	PU	M36
D9.2	Roadmap to TRL9 & Multi-Sector Adoption Strategy: Strategy for moving from TRL6 to TRL9, exploring cross-sector deployment, use-case integration and funding strategy and integrating stakeholders and end users feedback.	9	NORCE	R	PU	M48
D10.1	Review of Regulatory aspects of FC systems and Gap analysis	10	LR EMEA	R	PU	M9
D10.2	Risk assessment of fuel cell power system development	10	LR EMEA	R	PU	M15
D10.3	Risk assessment of modular arrangement of fuel cell power system	10	LR EMEA	R	PU	M36
D10.4	Safety analysis and Recommendations on FC Power systems	10	LR EMEA	R	PU	M45
D11.1	Plan for the communication, dissemination and exploitation of project results	11	DOWEL	R	SEN	M6
D12.1	Mapping and engagement activities towards standardisation bodies	12	Genevos	R	SEN	M48
D12.2	Pedagogical toolkit	12	MINES	R	PU	M48
D12.3	Final CDE report including the result and ownership list	12	DOWEL	R	SEN	M48

3.1.5 List of Milestones

Milestone Number and Name		WP	Date	Means of Verification
MS1	Preliminary FCS Architecture	3	M12	Final specification document approved by consortium;

	Defined			available in project repository
MS2	200 kW Prototype Assembled & Power-On Test	4	M23	Completion of build and initial test report; demonstration video or lab-recorded data
MS3	AST Protocol Validated	5	M24	Protocol documentation endorsed by partners; readiness for system stress test (review meeting)
MS4	Finalized test plan for 1 MW FCS approved	6	M29	Test plan is endorsed by partners and Energy House operators.
MS5	Fuel Cell System Installed and Commissioned at Energy House	6	M31	Integration and commissioning report of 1 MW FCS at Energy House is delivered and endorsed by partners.
MS6	Completion of 1,000-hour operational testing and decommissioning	6	M34	Test and decommissioning report is endorsed by partners and dataset interred in shared repository.
MS7	Digital-Twin Library Frozen for the demonstrations	7	M18	All PEMFC models needed for testing protocols and uploaded to shared repo; internal demo run recorded. Provides the “engine” for later HiL work.
MS8	HiL Software Stack Validated	7	M33	HiL test report showing closed-loop control of a virtual 1 MW FCS. Links WP7 → WP6.
MS9	LCA + TCO Draft Results	8	M38	Internal slide-deck with consolidated numbers for 1 MW & 10 MW cases distributed to partners; triggers WP9 scale-up discussion.
MS10	TRL-9 Roadmap Agreed	9	M44	Roadmap document endorsed by advisory board; identifies demo vessel, funding route, and certification actions.
MS11	Regulatory-Gap & Preliminary Safety Risk assessments	3, 10	M18	Regulatory-gap report (D10.1), Preliminary Risk Assessments (HAZID) on FC System (D10.2) and Modular systems, Draft FMEA (T3.4) report reviewed by LR EMEA + design team (VARD/GENEVOS). This green-lights component procurement.
MS12	Website online	11	M6	Availability of the project website on the internet
MS13	Identification of standardisation committees	12	M36	Internal documentation and presentation in a consortium meeting
MS14	Trainings organised	12	M48	Training minutes, D12.2

3.1.6 Critical Risks for Implementation

Description of risk / Likelihood / Severity				WP		Proposed risk-mitigation measures	
Delay in stack/component supply (e.g., supply chain disruptions affecting prototype build schedule).				M	M	4	Early engagement with suppliers and framework agreements, Identify backup suppliers, and monitor lead times via monthly procurement updates.
Hydrogen price increase: increased H2 costs reducing feasibility of test campaigns.				M	H	5,6	Budget for alternative test strategies (e.g., partial operation). Clarify risk with partners to allow scaling back of certain tasks if costs become prohibitive. Recycle produced energy to H2 production (SEC).
Inadequate FCS reliability under AST: system fails to meet durability targets in 40,000h accelerated test.				M	H	5	Conduct short-stack pre-testing to refine AST protocols; Allow extra iteration loop for design adjustments; Schedule mid-test checkpoints and corrections
Non-conformity between the 1 MW FCS and the Energy House test infrastructure (making integration more costly or unfeasible).				L	M	6	Repeat integration attempts with operational flexibility. Adjust testing-period to cover additional costs that occur due to an increase in costs for integration. The low severity is due to the strong experience of the involved partners in similar integration and test setups.
Safety / HSE incident (H2 leak, fire, high-pressure release) during module or 1 MW tests				L	H	5,6	Full HAZID/HAZOP & bow-tie analysis at M9 & M24; Drills & training, emergency response plan approved by LR EMEA; Insurance & contractual force-majeure clauses
Software/EMS integration failure (models ↔				M	M	5,6,	Adopt FMI standard wrappers, freeze comms spec at

controller ↔ plant)			7	M12; Fallback to manual load-sharing maps for demo.
Miss KPIs: on net efficiency (55 % LHV) or CAPEX ($\leq 1\,200$ €/kW)	M	M	3-6, 8	Iterative BoP optimisation using WP7 digital twin; Document gap & revise scale-up roadmap (WP9) showing cost-down lever and updated TRL path
Key partner insolvency or withdrawal	L	H	All	Activate MoUs with alternative labs/OEMs (identified by WP1)
Budget over-run (CAPEX or OPEX)	M	M	1,2	Early warnings in MC meetings; Re-prioritise optional tasks; seek in-kind contributions
Data/cyber incident jeopardising open-source release or IP leak	L	M	7, 11	Version-controlled repo with access tiers; Encrypted backups; GDPR/IPR compliance checks; Isolate breach, regenerate repos; legal follow-up.

3.1.7 Summary of Staff Effort

Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP10	WP11	WP12	Total
1-NORCE	12,0	12,0	3,0			2,5			19,2	2,0	1,0	1,0	52,7
2-ARMINES							18,0						18,0
3-MINES	1,0	0,5	5,3		5,0		16,0				0,5	3,0	31,3
4-VARD	0,5	0,5	9,0	3,5	2,5	6,0	5,2	3,6	6,5	4,5	0,5	1,0	43,3
5-GENEVOS	0,5	0,5	39,8	12,6	4,5	20,0			1,0	2,0	0,5	1,9	83,3
6-USTUTT	0,5	0,5					34,0				0,5	0,5	36,0
7-LR EMEA	1,0	1,0	0,5	0,5	1,5	1,0	1,0	1,0	1,0	13,5	1,0	1,0	24,0
8-CERTH	1,0	1,0						25,5	8,0		1,0	1,0	37,5
9-DOWEL	1,0	1,0									10,0	12,5	24,5
10-SEC	0,5	0,5				9,0					0,5	0,5	11,0
11-CEA	1,0	1,0	2,5	5,0	25,5	2,0	3,0		2,0		0,5	0,5	43,0
12-KIARA	0,4	0,6						1,5	4,5		0,5	0,5	8,0
13-NJORD								1,0	1,0	1,0			3,0
14-SCORPIO (AE)									4,8				4,8
TOTAL PMs	19,4	19,1	60,1	21,6	39,0	40,5	77,2	32,6	48,0	23,0	16,5	23,4	420,4

3.1.8 Subcontracting

Participant	Cost (€)	Description of tasks and justification
5-GENEVOS	100 000	WP3: Engineering support from external suppliers: Fuel cell stack, H2 pump, compressor, cooling, which are only available commercially and externally to the consortium

3.1.9 Purchase costs

2-ARMINES	Cost (€)	Justification
Travel and subsistence	17 000	17 travels, assuming 1k€ per traveller per trip. Trips include 2 consortium and review meetings in WP1&WP2; 2 participants in 1 WP3 technical meeting; 2 trips for co-supervision with CEA in WP5; 2-3 participants in 3 WP7 technical meetings; 1 trip related to training on modelling fuel cell systems as well as 2 conference travels in WP11&WP12.
Other goods, works and services	24 570	WP3: Purchase of 1 short stack (5000€) for early stack performance characterization for BoP architecture definition. Stack bench utilisation (5220€): Early stack performance characterization for BoP architecture definition. 60 days of tests at 87 EUR/day (internal estimation based on running costs) for stack setup, performance and lifetime characterization, 8 h/day. WP5: Purchase of 2 short-stacks (10000€) for model calibration and validation of AST protocols before 200 kW tests. Stack bench utilisation (4350€): Running 400h for AST tests, 50 days of tests.
Remaining purchase costs (<15% of pers.costs)	7 370	
Total	48 940	

5-GENEVOS	Cost (€)	Justification
Travel and subsistence	90 400	WP3: Travel in WP3-WP6 for 90 days each year/360 travel days over the 4-year project. The costs include 360 hotel nights at 120€ per night; rental car for 180 days, 120€ per day, transport costs, 6400€ per year for 4 years.
Other goods, works and services	797 900	WP4: Purchase of various consumables for the two systems, 200/250 kW FCS for AST (A-Proto) and 1 MW FCS (B-Proto), incl. fuel cell stacks (346144€), air loop (165894€), enclosure/mechanical/interfaces (129004€), H2 loop (102655€), coolant loop (54203€).
Remaining purchase costs (<15% of pers.costs)	87 307	
Total	975 607	

8-CERTH	Cost (€)	Justification
Other goods, works and services	27 200	WP8: 2 integrated licences for LCA assessment at 3000€ each per year for 3 years, and add-in for softwares for integrated solutions, at 1050€ per year for 2 licences for 2 years. WP11: Organization of 1st workshop together with Greek hub stakeholders and networking with other projects, 5000€.
Remaining purchase costs (<15% of pers.costs)	21 200	
Total	48 400	

10-SEC	Cost (€)	Justification
Other goods, works and services	525 024	WP6: Installation and decommissioning of containerized fuel cell solution at Energy House. Estimated cost: Materials and equipment 99311€ (prep of test facility, telecom system, lighting system, HVAC, piping, instrument air, electrical distr. panel, foundations instrumentation, F&G and SD system); installation hours 78762€ , engineering hours 73038€ (client meetings, lead engineering, block interface design, 3D modelling, piping ISO, electrical design, etc.). Purchase of 30 tons of H2 at an estimated price of 9,545€ per kg H2. Calculation is based on 1000 hours of running the fuel cell, with an average load of 80 % for 9 hours a day, and 50 % load for 15 hours a day, with continuous testing in 42 days.
Remaining purchase costs (<15% of pers.costs)	17 195	
Total	524 219	

11-CEA	Cost (€)	Justification
Other goods, works and services	140 000	WP5: Purchase of approx 8.75 tons of H2 at an estimated price of 16€ per KG of H2.
Remaining purchase costs (<15% of pers. costs)	33 700	
Total	173 700	

12-KIARA	Cost (€)	Justification
Travel and subsistence	9 000	WP1&WP2: Participation in 4 consortium meetings, incl. review meeting, closing meeting. Cost per traveller 750€ (estimate assumes 1 person for 1 night, medium haul Intra European travel), 2 travellers per trip. WP9: Participating in meeting with stakeholders. 2 meetings, 1

		participant per meeting at 1500€ per trip.
Other goods, works and services	8 000	WP11&WP12: Organization of workshop for policy makers (5000€) and organization of workshop together with Greek hub stakeholders (3000€).
Remaining purchase costs (<15% of pers. costs)	2 000	
Total	19 000	

3.1.10 Other costs categories

10-SEC	Cost (€)	Justification
Internally invoiced goods and services	103 941	WP6: Testing of 1 MW fuel cell. Costs include day rent of test cell 84120€, rent of switchboard 5380€, facility costs 7189€ and preliminary costs 7251€. Covers a period of 100 days from installation to complete decommissioning, based on 42 days of continuous testing.
11-CEA	Cost (€)	Justification
Internally invoiced goods and services	197 380	WP3-WP7&WP9: Platform costs (internally invoiced goods and services). In all projects, CEA uses specific platforms to carry out the technical tasks described in the WPs. In this project, CEA will use the SEANERGY platform. The costs are calculated by multiplying the hours worked on the project by the hourly rate in accordance with HE rules (Article 6.2.D.2 of HE Model Grant Agreement). For this project, CEA will rely on the SEANERGY platform for a total of 5113.5 hours @38.60€/h in WP3 to 7 and 9 for FC testing and modelling activities.

##\$QUA-LIT-QL\$# ##\$WRK-PLA-WP\$#

3.2 Capacity of participants and consortium as a whole ##@CON-SOR-CS@## @PRJ-MGT-PM@#

This consortium unites a complementary set of actors from research and academia (NORCE, ARMINES/MINES, USTUTT, CERTH), industry (GENEVOS, VARD, LR EMEA, and SCORPIO), as well as innovation and communication specialists (Dowel), and demonstration providers (CEA, Sustainable Energy). As an OEM of fuel cell systems, Genevos ensures direct technology development and integration capabilities while VARD has proven expertise in multi-MW hybrid, diesel-electric, and full-electric configurations. This combination makes the consortium well positioned to manage the technical challenges of multi-stack fuel cell system integration and demonstration. The end users KIARA (SEAJETS), SCORPIO, and NJORD bring real-world maritime needs and operational feedback covering the key areas of shipping and marine transportation. Test labs and demonstration facilities, provided by Sustainable Energy, CEA and supported by the RTOs/universities, enable practical validation under realistic conditions. LR EMEA contributes essential classification and certification expertise, ensuring that solutions align with maritime regulations and safety standards. Meanwhile, the research organizations offer deep technical knowledge in control software, simulation, life-cycle assessment (LCA), total cost of ownership (TCO) evaluation, and advanced materials/durability testing. This combination ensures that each step in the value chain, from innovation, prototyping, and demonstration, to market alignment, and regulatory acceptance, is robustly addressed for the successful delivery of a scalable multi-MW fuel cell system.



NORCE is a Norwegian independent research institute with a particular focus on applied R&D in energy, and technology with strong connections to the maritime industry. The NORCE team includes the group formerly known as CLARA VENTURE LABS. **Main Role:** Project coordinator and oversees administrative, financial, and risk management. Facilitates synergy among partners and ensures timely progress across work packages. Leader of WP on Scale-Up, & End-user engagement.

ARMINES-MINES A top French engineering school known for high-level research in energy systems, automation,

  	and sustainability, with 30 years of experience in fuel cell research. Main Role: Develops advanced physics-based models, monitoring and control algorithms, contributes to the definition of AST protocols for the FCS.
	VARD Electro A fully owned subsidiary of VARD Group, is a leading system integrator delivering turnkey electrical systems and solutions for the global maritime industry. With a deep knowledge in integration and controlling of marine power systems, both AC and DC solution, focusing on reduction of consumption and emission.
	GENEVOS A French OEM focusing on robust and scalable PEM fuel cell systems for maritime applications. Main Role: Leads FCS design and assembly, integrating multi-stack solutions for the 1–10 MW scale while ensuring cost-effectiveness and marine suitability.
 University of Stuttgart Institute of Automotive Engineering	USTUTT , more specifically its Institute of Automotive Engineering, is a public research institute with deep expertise in automotive powertrain systems, control software, application of Machine Learning methods to engineering problems and engineering software development Main Role: USTUTT is responsible for the integration of the physics- and data-based models developed within the project into open-source software and its publication.
	LR EMEA A globally recognized maritime classification society offering compliance, certification, and risk assessments. Main Role: Leads WP on Provides regulatory guidance, conducts risk evaluations, and ensures the fuel cell system adheres to maritime standards and type-approval processes.
	SCORPIO , A major international shipping company based in Monaco, managing over 130 vessels including product and gas tankers, with a strong focus on safety, protection of environment in general, decarbonization and operational efficiency. Main Role: Provides vessel-specific operational data across multiple ship types (e.g., product and gas tankers, dry bulkers, container ships, wind turbine installation vessels). Contributes end-user feedback on safety, crew training, and integration requirements and co-develops a use case to support scaling from TRL6 to TRL9. SCORPIO participates as a self-financed Affiliated partner from outside the EU. No ethical issues are expected because of this.
 CERTH CENTRE FOR RESEARCH & TECHNOLOGY HELLAS	CERTH Greece's foremost multidisciplinary research center, with strong expertise in techno-economic analyses, LCA, and sustainability and experience with leading EU projects in Hydrogen research. Main Role: leads the techno-economic and life cycle assessments. Will organize workshops with the H2MARINE team ensuring complementarity, and to leverage synergies and avoid duplication of efforts.
	DOWEL A French consultancy specializing in communication, dissemination, and knowledge transfer for European research projects. Main Role: Manages the project's communication plan, stakeholder engagement, exploitation strategy, and open science activities to maximize impact.
	SEC A Norwegian test center focusing on hydrogen technologies, providing testing and demonstration facilities for clean energy solutions. Main Role: Leads the WP on the 1 MW FCS test and demonstration under maritime-like conditions to validate system performance, reliability, and durability.
	CEA is a public Research and Technology Organisation (RTO). With a staff of about 21,000 employees, over 650 priority patents per year and more than 200 high-tech spin-offs since its creation, CEA is the largest technology research and innovation provider in France. Main Role: Leads the WP on, and will perform the accelerated stress tests for the 200 kW sub-system, and develop the new protocols for these.
	KIARA is a company operating under the brand name of the SEAJETS group, a major Greek ferry operator with the world's largest high-speed catamaran fleet, serving routes across the Aegean Sea. Main Role: Provides operational data and a real and detailed use case for a high-speed passenger ferry in the Aegean. Offers end-user feedback on safety, integration, and crew training.
	NJORD is a Copenhagen-based maritime decarbonization firm, founded by Maersk Tankers, specializing in integrating and scaling energy efficiency technologies across global fleets. Main Role: Provides expertise in evaluating end user business cases, supporting the consortium with operational experience, and contributing to defining realistic vessel integration pathways, safety practices, and scalability considerations for containerized maritime hydrogen solutions.

#§CON-SOR-CS§# #§PRJ-MGT-PM§#